

Concept 1 | Complex Numbers

English Student Edition • Focused formulas and guided practice

Formula opener

Standard form

$$z = a + bi$$

Conjugate

$$\overline{a + bi} = a - bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Worked Solutions

Complex Numbers

Q001 Written

State the real and imaginary parts of $3 - 2i$.**Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(3, -2)$.

Final answer $(3, -2)$

Q002 Written

State the real and imaginary parts of $(4 - \sqrt{5}i)/3$.**Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(4/3, -\sqrt{5}/3)$.

Final answer $(4/3, -\sqrt{5}/3)$

Worked Solutions

Complex Numbers

Q003 Written

State the real and imaginary parts of -8 .

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-8, 0)$.

Final answer

$(-8, 0)$

Q004 Written

State the real and imaginary parts of $2i$.

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0, 2)$.

Final answer

$(0, 2)$

Worked Solutions

Complex Numbers

Q005 Written

Find the real values of x and y satisfying $(2 + i)x + (1 + i)y = 2 + 3i$, x, y real.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = -1, y = 4$.

Final answer

$$x = -1, y = 4$$

Q006 Written

State the real and imaginary parts of $-3 + 5i$.

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-3, 5)$.

Final answer

$$(-3, 5)$$

Worked Solutions

Complex Numbers

Q007 Written

State the real and imaginary parts of $(-1 - \sqrt{3}i)/2$.

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-1/2, -\sqrt{3}/2)$.

Final answer $(-1/2, -\sqrt{3}/2)$

Q008 Written

State the real and imaginary parts of 1.

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(1, 0)$.

Final answer $(1, 0)$

Worked Solutions

Complex Numbers

Q009 Written

State the real and imaginary parts of $3i$.**Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0, 3)$.

Final answer $(0, 3)$

Q010 Written

Find the real values of x and y satisfying $(x - 6) + (x + 3y)i = 0$, x, y real.**Solution**

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 6, y = -2$.

Final answer $x = 6, y = -2$

Worked Solutions

Complex Numbers

Q011 Written

Find the real values of x and y satisfying $(3 + 2i)x + (4 - i)y = 6 - 7i$, x, y real.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = -2, y = 3$.

Final answer

$$x = -2, y = 3$$

Q012 Written

State the real and imaginary parts of $2 + 5i$.

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(2, 5)$.

Final answer

$$(2, 5)$$

Worked Solutions

Complex Numbers

Q013 Written

State the real and imaginary parts of $-25 + 13i$.

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-25, 13)$.

Final answer

$(-25, 13)$

Q014 Written

State the real and imaginary parts of $(2 - \sqrt{5}i)/3$.

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(2/3, -\sqrt{5}/3)$.

Final answer

$(2/3, -\sqrt{5}/3)$

Worked Solutions

Complex Numbers

Q015 Written

State the real and imaginary parts of $(7 - \sqrt{2}i)/2$.

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(7/2, -\sqrt{2}/2)$.

Final answer $(7/2, -\sqrt{2}/2)$

Q016 Written

State the real and imaginary parts of -4 .

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-4, 0)$.

Final answer $(-4, 0)$

Worked Solutions

Complex Numbers

Q017 Written

State the real and imaginary parts of $1/4$.**Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(1/4, 0)$.

Final answer $(1/4, 0)$

Q018 Written

State the real and imaginary parts of $\sqrt{3}i$.**Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0, \sqrt{3})$.

Final answer $(0, \sqrt{3})$

Worked Solutions

Complex Numbers

Q019 Written

State the real and imaginary parts of $-i$.**Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0, -1)$.

Final answer $(0, -1)$

Q020 Written

Find the real values of x and y satisfying $(2x + y) + (3x - y)i = 8 + 7i$.**Solution**

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 3, y = 2$.

Final answer $x = 3, y = 2$

Worked Solutions

Complex Numbers

Q021 Written

Find the real values of x and y satisfying $(x + 1) - 5i = 4 + (2y + 1)i$.**Solution**

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 3, y = -3$.

Final answer

$$x = 3, y = -3$$

Q022 Written

Find the real values of x and y satisfying $(x - 2) + (x + y)i = 0$.**Solution**

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 2, y = -2$.

Final answer

$$x = 2, y = -2$$

Worked Solutions

Complex Numbers

Q023 Written

Find the real values of x and y satisfying $(3 + 2i)x + (1 - i)y = 7 + 3i$.**Solution**

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 2, y = 1$.

Final answer

$$x = 2, y = 1$$

Q024 Written

Find the real values of x and y satisfying $(1 - i)(x + yi) = 2 + i$.**Solution**

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 1/2, y = 3/2$.

Final answer

$$x = 1/2, y = 3/2$$

Worked Solutions

Complex Numbers

Q025 Written

Solve the complex-number equation and find the exact real values of x and y : $(1 + 2i)(x + i) = x(y + i)$.

Solution

1. Expand: $((1 + 2i)(x + i) = x - 2 + (1 + 2x)i)$.
2. Equate coefficients with $(xy + xi)$: $(x - 2 = xy)$ and $(1 + 2x = x)$.
3. Solving gives $x = -1, y = 3$.

Final answer

$$x = -1, y = 3$$

Q026 Written

Calculate $(3 - 2i) - (2 + 5i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $1 - 7i$.

Final answer

$$1 - 7i$$

Worked Solutions

Complex Numbers

Q027 **Written****Calculate $(i - 2)^2$ and give the answer in standard form.****Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $3 - 4i$.

Final answer

$$3 - 4i$$

Q028 **Written****Calculate $i^6 + i^{27} + i^{17}$ and give the answer in standard form.****Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: -1 .

Final answer

$$-1$$

Worked Solutions

Complex Numbers

Q029 Written

Calculate $(4 + 6i) + (-3 + 2i)$ and give the answer in standard form.**Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $1 + 8i$.

Final answer

$$1 + 8i$$

Q030 Written

Calculate $(2 + 3i)(1 - 2i)$ and give the answer in standard form.**Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $8 - i$.

Final answer

$$8 - i$$

Worked Solutions

Complex Numbers

Q031 Written

Calculate $i + i^2 + i^3 + i^4$ and give the answer in standard form.**Solution**

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: 0.

Final answer

0

Q032 Written

Calculate $(7 + 4i) - (3 + 5i)$ and give the answer in standard form.**Solution**

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $4 - i$.

Final answer $4 - i$

Worked Solutions

Complex Numbers

Q033 Written

Calculate $(-4 + 3i) + (5 - 6i)$ and give the answer in standard form.**Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $1 - 3i$.

Final answer

$$1 - 3i$$

Q034 Written

Calculate $(3 - 3i) - (5 - 3i)$ and give the answer in standard form.**Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: -2 .

Final answer

$$-2$$

Worked Solutions

Complex Numbers

Q035 **Written****Calculate $(\sqrt{2} + 3i)(\sqrt{2} + i)$ and give the answer in standard form.****Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $-1 + 4\sqrt{2}i$.

Final answer

$$-1 + 4\sqrt{2}i$$

Q036 **Written****Calculate $(2 + 3i) + (5 + 8i)$ and give the answer in standard form.****Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $7 + 11i$.

Final answer

$$7 + 11i$$

Worked Solutions

Complex Numbers

Q037 Written

Calculate $(2/5 - 1/2i) - (1/3 + 3/7i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $1/15 - 13/14i$.

Final answer

$$1/15 - 13/14i$$

Q038 Written

Calculate $(2i - 1)^2$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $-3 - 4i$.

Final answer

$$-3 - 4i$$

Worked Solutions

Complex Numbers

Q039 Written

Calculate $(2/i + i)(2/i - i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: -3 .

Final answer -3

Q040 Written

Calculate $(2 + i)^3 + (2 - i)^3$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: 4.

Final answer 4

Worked Solutions

Complex Numbers

Q041 Written

Calculate $i^5 + i^3 + i$ and give the answer in standard form.**Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: i .

Final answer i

Q042 Written

Calculate $i^9 + i^{14} + i^{11}$ and give the answer in standard form.**Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: -1 .

Final answer -1

Worked Solutions

Complex Numbers

Q043 Written

Calculate $i^{12} + i^{18} + i^{15}$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $-i$.

Final answer $-i$

Q044 Written

Given x is real, solve $(7x - 7i)(x^2 + xi) = 14x^2i^{40}$.

Solution

1. Use $(i^{40} = 1)$ and $((x - i)(x + i) = x^2 + 1)$.
2. The equation becomes $(7x(x^2 + 1) = 14x^2)$, so $(7x(x - 1)^2 = 0)$.
3. Therefore $x = 0$ or $x = 1$.

Final answer $x = 0$ or $x = 1$

Worked Solutions

Complex Numbers

Q045 Written

State the complex conjugate of $2 + 3i$.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $2 - 3i$.

Final answer

$$2 - 3i$$

Q046 Written

State the complex conjugate of $7 - i\sqrt{11}$.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $7 + i\sqrt{11}$.

Final answer

$$7 + i\sqrt{11}$$

Worked Solutions

Complex Numbers

Q047 **Written****State the complex conjugate of 6.****Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is 6.

Final answer

6

Q048 **Written****State the complex conjugate of $-5i$.****Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $5i$.

Final answer $5i$

Worked Solutions

Complex Numbers

Q049 Written

Calculate $(2 + i)/(2 - i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $3/5 + 4/5i$.

Final answer

$$3/5 + 4/5i$$

Q050 Written

Calculate $(3 - i)/(2i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $-1/2 - 3/2i$.

Final answer

$$-1/2 - 3/2i$$

Worked Solutions

Complex Numbers

Q051 **Written****State the complex conjugate of $1 - 4i$.****Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $1 + 4i$.

Final answer

$$1 + 4i$$

Q052 **Written****State the complex conjugate of $\sqrt{2} + i\sqrt{10}$.****Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $\sqrt{2} - i\sqrt{10}$.

Final answer

$$\sqrt{2} - i\sqrt{10}$$

Worked Solutions

Complex Numbers

Q053 Written

State the complex conjugate of 2.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is 2.

Final answer

2

Q054 Written

State the complex conjugate of $-8i$.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $8i$.

Final answer $8i$

Worked Solutions

Complex Numbers

Q055 Written

Calculate $(3 + i)/(1 + 2i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $1 - i$.

Final answer

$$1 - i$$

Q056 Written

Calculate $(1 - i)/i$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $-1 - i$.

Final answer

$$-1 - i$$

Worked Solutions

Complex Numbers

Q057 **Written****State the complex conjugate of $3 + 4i$.****Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $3 - 4i$.

Final answer

$$3 - 4i$$

Q058 **Written****State the complex conjugate of $9 + 8i$.****Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $9 - 8i$.

Final answer

$$9 - 8i$$

Worked Solutions

Complex Numbers

Q059 Written

State the complex conjugate of $8 + i\sqrt{3}$.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $8 - i\sqrt{3}$.

Final answer

$$8 - i\sqrt{3}$$

Q060 Written

State the complex conjugate of 5.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is 5.

Final answer

$$5$$

Worked Solutions

Complex Numbers

Q061 Written

State the complex conjugate of -15 .**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is -15 .

Final answer -15

Q062 Written

State the complex conjugate of $\sqrt{7}$.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $\sqrt{7}$.

Final answer $\sqrt{7}$

Worked Solutions

Complex Numbers

Q063 Written

State the complex conjugate of i .**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $-i$.

Final answer $-i$

Q064 Written

State the complex conjugate of $-50i$.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $50i$.

Final answer $50i$

Worked Solutions

Complex Numbers

Q065 Written

State the complex conjugate of $i\sqrt{6}$.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $-i\sqrt{6}$.

Final answer

$$-i\sqrt{6}$$

Q066 Written

Calculate $(1 + i)/(1 - i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is i .

Final answer

$$i$$

Worked Solutions

Complex Numbers

Q067 Written

Calculate $(3 + i)/(2 + i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $7/5 - 1/5i$.

Final answer

$$7/5 - 1/5i$$

Q068 Written

Calculate $(\sqrt{5} + i)/(\sqrt{5} - i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $2/3 + \sqrt{5}/3i$.

Final answer

$$2/3 + \sqrt{5}/3i$$

Worked Solutions

Complex Numbers

Q069 Written

Calculate $5i/(1 + 2i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $2 + i$.

Final answer

$$2 + i$$

Q070 Written

Calculate $5i/(4 + 3i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $3/5 + 4/5i$.

Final answer

$$3/5 + 4/5i$$

Worked Solutions

Complex Numbers

Q071 Written

Calculate $(1 - 3i)/(3 - 2i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $9/13 - 7/13i$.

Final answer

$$9/13 - 7/13i$$

Q072 Written

Calculate $1/i$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $-i$.

Final answer

$$-i$$

Worked Solutions

Complex Numbers

Q073 Written

Calculate $(2 + i)/(3i)$ and give the answer in the form $a+bi$.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $1/3 - 2/3i$.

Final answer

$$1/3 - 2/3i$$

Q074 Written

Simplify exactly: $\left(\frac{i(10-12i)}{(2+i)(-1+4i)} \right)^{2027}$.

Solution

1. Rationalise the base using the conjugate of the denominator.
2. The base simplifies exactly to $(-2-144i)/85$.
3. Hence the exact result is $((-2 - 144i)/85)^{2027}$; no expansion of the 2027th power is required.

Final answer

$$((-2 - 144i)/85)^{2027}$$

Worked Solutions

Complex Numbers

Q075 Written

Express the square roots of -6 in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $\pm i\sqrt{6}$.

Final answer

$$\pm i\sqrt{6}$$

Q076 Written

Express $\sqrt{-12}\sqrt{-6}$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-6\sqrt{2}$.

Final answer

$$-6\sqrt{2}$$

Worked Solutions

Complex Numbers

Q077 Written

Express $\sqrt{8}/\sqrt{-18}$ in terms of i .**Solution**

1. Use $(\sqrt{a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-(2/3)i$.

Final answer

$$-(2/3)i$$

Q078 Written

Express $\sqrt{-7}$ in terms of i .**Solution**

1. Use $(\sqrt{a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $i\sqrt{7}$.

Final answer

$$i\sqrt{7}$$

Worked Solutions

Complex Numbers

Q079 Written

Express $\sqrt{-27}$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $3i\sqrt{3}$.

Final answer

$$3i\sqrt{3}$$

Q080 Written

Express $-\sqrt{-12}$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-2i\sqrt{3}$.

Final answer

$$-2i\sqrt{3}$$

Worked Solutions

Complex Numbers

Q081 Written

Express the square roots of -12 in terms of i .**Solution**

1. Use ($\sqrt{-a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $\pm 2i\sqrt{3}$.

Final answer

$$\pm 2i\sqrt{3}$$

Q082 Written

Express $\sqrt{-3}\sqrt{-6}$ in terms of i .**Solution**

1. Use ($\sqrt{-a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $-3\sqrt{2}$.

Final answer

$$-3\sqrt{2}$$

Worked Solutions

Complex Numbers

Q083 Written

Express $\sqrt{27}/\sqrt{-3}$ in terms of i .**Solution**

1. Use $(\sqrt{a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-3i$.

Final answer $-3i$

Q084 Written

Express $(3 + \sqrt{-2})^2$ in terms of i .**Solution**

1. Use $(\sqrt{a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $7 + 6\sqrt{2}i$.

Final answer $7 + 6\sqrt{2}i$

Worked Solutions

Complex Numbers

Q085 Written

Express $\sqrt{-2}$ in terms of i .**Solution**

1. Use ($\sqrt{a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $i\sqrt{2}$.

Final answer

$$i\sqrt{2}$$

Q086 Written

Express $\sqrt{-9}$ in terms of i .**Solution**

1. Use ($\sqrt{a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $3i$.

Final answer

$$3i$$

Worked Solutions

Complex Numbers

Q087 Written

Express $\sqrt{-45}$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $3i\sqrt{5}$.

Final answer

$$3i\sqrt{5}$$

Q088 Written

Express $-\sqrt{-24}$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-2i\sqrt{6}$.

Final answer

$$-2i\sqrt{6}$$

Worked Solutions

Complex Numbers

Q089 Written

Express $-\sqrt{-36}$ in terms of i .**Solution**

1. Use $(\sqrt{a})^2 = a$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-6i$.

Final answer $-6i$

Q090 Written

Express the square roots of -2 in terms of i .**Solution**

1. Use $(\sqrt{a})^2 = a$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $\pm i\sqrt{2}$.

Final answer $\pm i\sqrt{2}$

Worked Solutions

Complex Numbers

Q091 Written

Express the square roots of -18 in terms of i .**Solution**

1. Use ($\sqrt{-a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $\pm 3i\sqrt{2}$.

Final answer

$$\pm 3i\sqrt{2}$$

Q092 Written

Express the square roots of -16 in terms of i .**Solution**

1. Use ($\sqrt{-a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $\pm 4i$.

Final answer

$$\pm 4i$$

Worked Solutions

Complex Numbers

Q093 Written

Express $\sqrt{-2}\sqrt{-6}$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-2\sqrt{3}$.

Final answer

$$-2\sqrt{3}$$

Q094 Written

Express $\sqrt{-28}\sqrt{-35}$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-14\sqrt{5}$.

Final answer

$$-14\sqrt{5}$$

Worked Solutions

Complex Numbers

Q095 Written

Express $\sqrt{8}/\sqrt{-2}$ in terms of i .**Solution**

1. Use $(\sqrt{a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-2i$.

Final answer $-2i$

Q096 Written

Express $\sqrt{27}/\sqrt{-12}$ in terms of i .**Solution**

1. Use $(\sqrt{a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-(3/2)i$.

Final answer $-(3/2)i$

Worked Solutions

Complex Numbers

Q097 Written

Express $\sqrt{-32} + \sqrt{-98} + \sqrt{-8}$ in terms of i .**Solution**

1. Use ($\sqrt{a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $13i\sqrt{2}$.

Final answer

$$13i\sqrt{2}$$

Q098 Written

Express $\sqrt{-98} - \sqrt{-72} + \sqrt{-50}$ in terms of i .**Solution**

1. Use ($\sqrt{a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $6i\sqrt{2}$.

Final answer

$$6i\sqrt{2}$$

Worked Solutions

Complex Numbers

Q099 Written

Express $(\sqrt{3} + \sqrt{-5})^2$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-2 + 2i\sqrt{15}$.

Final answer

$$-2 + 2i\sqrt{15}$$

Q100 Written

Express $(2 + \sqrt{-3})(2 - \sqrt{-3})$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore 7.

Final answer

$$7$$

Worked Solutions

Complex Numbers

Q101 Written

Solve over the complex numbers: $(2x - 8)^2 = -10$.

Solution

1. Take square roots: $(2x - 8 = \pm \sqrt{10})$.
2. Divide by 2 to obtain $x = 4 \pm (\sqrt{10}/2)i$.

Final answer

$$x = 4 \pm (\sqrt{10}/2)i$$

Q102 MCQ

For $2 + 3i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $2 + 3i$ using the relevant family rule.
2. The correct value is (2, 3).

Correct option: C

Final answer

(2, 3)

Worked Solutions

Complex Numbers

Q103 MCQ

For $-4 + 5i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-4 + 5i$ using the relevant family rule.
2. The correct value is $(-4, 5)$.

Correct option: D

Final answer

$(-4, 5)$

Q104 MCQ

For $7 - 2i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $7 - 2i$ using the relevant family rule.
2. The correct value is $(7, -2)$.

Correct option: D

Final answer

$(7, -2)$

Worked Solutions

Complex Numbers

Q105 MCQ

For $(1 - \sqrt{2}i)/3$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $(1 - \sqrt{2}i)/3$ using the relevant family rule.

2. The correct value is $\left(\frac{1}{3}, -\frac{\sqrt{2}}{3}\right)$.

Correct option: B

Final answer

$$\left(\frac{1}{3}, -\frac{\sqrt{2}}{3}\right)$$

Q106 MCQ

For -6 , select the ordered pair (real part, imaginary part):

Solution

1. Simplify -6 using the relevant family rule.

2. The correct value is $(-6, 0)$.

Correct option: D

Final answer

$$(-6, 0)$$

Worked Solutions

Complex Numbers

Q107 MCQ

For $4i$, select the ordered pair (real part, imaginary part):**Solution**

1. Simplify $4i$ using the relevant family rule.
2. The correct value is $(0, 4)$.

Correct option: A**Final answer** $(0, 4)$

Q108 MCQ

For $-i$, select the ordered pair (real part, imaginary part):**Solution**

1. Simplify $-i$ using the relevant family rule.
2. The correct value is $(0, -1)$.

Correct option: B**Final answer** $(0, -1)$

Worked Solutions

Complex Numbers

Q109 MCQ

For $\sqrt{5}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $\sqrt{5}i$ using the relevant family rule.
2. The correct value is $(0, \sqrt{5})$.

Correct option: C**Final answer** $(0, \sqrt{5})$

Q110 MCQ

For $3/2 + \sqrt{3}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $3/2 + \sqrt{3}i$ using the relevant family rule.
2. The correct value is $(\frac{3}{2}, \sqrt{3})$.

Correct option: B**Final answer** $(\frac{3}{2}, \sqrt{3})$

Worked Solutions

Complex Numbers

Q111 MCQ

For $-1/4 - 2i$, select the ordered pair (real part, imaginary part):**Solution**

1. Simplify $-1/4 - 2i$ using the relevant family rule.
2. The correct value is $\left(-\frac{1}{4}, -2\right)$.

Correct option: D**Final answer**

$$\left(-\frac{1}{4}, -2\right)$$

Q112 MCQ

For $5/3$, select the ordered pair (real part, imaginary part):**Solution**

1. Simplify $5/3$ using the relevant family rule.
2. The correct value is $\left(\frac{5}{3}, 0\right)$.

Correct option: D**Final answer**

$$\left(\frac{5}{3}, 0\right)$$

Worked Solutions

Complex Numbers

Q113 MCQ

For $-\sqrt{7}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-\sqrt{7}i$ using the relevant family rule.
2. The correct value is $(0, -\sqrt{7})$.

Correct option: C

Final answer

$(0, -\sqrt{7})$

Q114 MCQ

For $-9 + \sqrt{2}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-9 + \sqrt{2}i$ using the relevant family rule.
2. The correct value is $(-9, \sqrt{2})$.

Correct option: D

Final answer

$(-9, \sqrt{2})$

Worked Solutions

Complex Numbers

Q115 MCQ

For $2/5 - 3/4i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $2/5 - 3/4i$ using the relevant family rule.
2. The correct value is $\left(\frac{2}{5}, -\frac{3}{4}\right)$.

Correct option: B

Final answer

$$\left(\frac{2}{5}, -\frac{3}{4}\right)$$

Q116 MCQ

For $-\sqrt{11}$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-\sqrt{11}$ using the relevant family rule.
2. The correct value is $(-\sqrt{11}, 0)$.

Correct option: B

Final answer

$$(-\sqrt{11}, 0)$$

Worked Solutions

Complex Numbers

Q117 MCQ

For $\sqrt{13}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $\sqrt{13}i$ using the relevant family rule.
2. The correct value is $(0, \sqrt{13})$.

Correct option: C**Final answer** $(0, \sqrt{13})$

Q118 MCQ

For $(2 + i)x + (1 - i)y = 3 + 3i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(2, -1)$.

Correct option: D**Final answer** $(2, -1)$

Worked Solutions

Complex Numbers

Q119 MCQ

For $(1 + 2i)x + (3 - i)y = 5 - 4i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(-1, 2)$.

Correct option: A

Final answer $(-1, 2)$

Q120 MCQ

For $(3 - 2i)x + (1 + i)y = 6 + i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(1, 3)$.

Correct option: B

Final answer $(1, 3)$

Worked Solutions

Complex Numbers

Q121 MCQ

For $(2 - i)x + (4 + 2i)y = 4i$, select the ordered pair (x, y) :

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(-2, 1)$.

Correct option: B

Final answer

$(-2, 1)$

Q122 MCQ

For $(1 + i)x + (2 + 3i)y = 1$, select the ordered pair (x, y) :

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(3, -1)$.

Correct option: C

Final answer

$(3, -1)$

Worked Solutions

Complex Numbers

Q123 MCQ

For $(4 + i)x + (1 - 2i)y = 2 + 5i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(1, -2)$.

Correct option: D

Final answer

$(1, -2)$

Q124 MCQ

For $(2 + 3i)x + (3 + i)y = -8 - 5i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(-1, -2)$.

Correct option: D

Final answer

$(-1, -2)$

Worked Solutions

Complex Numbers

Q125 MCQ

For $(1 - i)x + (2 - i)y = 6 - 4i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives (2, 2).

Correct option: C

Final answer

(2, 2)

Q126 MCQ

For $(3 + 4i) - (1 - 2i)$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $2 + 6i$.

Correct option: C

Final answer

$2 + 6i$

Worked Solutions

Complex Numbers

Q127 MCQ

For $(-2 + i) + (5 - 3i)$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $3 - 2i$.

Correct option: C**Final answer**

$$3 - 2i$$

Q128 MCQ

For $(1 + 2i)(3 - i)$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $5 + 5i$.

Correct option: D**Final answer**

$$5 + 5i$$

Worked Solutions

Complex Numbers

Q129 MCQ

For $(\sqrt{3} + i)(\sqrt{3} - i)$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is 4.

Correct option: A

Final answer

4

Q130 MCQ

For $(2 - i)^2$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $(2 - i)^2$.

Correct option: C

Final answer $(2 - i)^2$

Worked Solutions

Complex Numbers

Q131 MCQ

For $(1 + 3i)^2$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-8 + 6i$.

Correct option: A**Final answer** $-8 + 6i$

Q132 MCQ

For i^{37} , select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is i .

Correct option: B**Final answer** i

Worked Solutions

Complex Numbers

Q133 MCQ

For $i^{22} + i^9$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-1 + i$.

Correct option: B**Final answer**

$$-1 + i$$

Q134 MCQ

For $i^{15} - i^6$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $1 - i$.

Correct option: B**Final answer**

$$1 - i$$

Worked Solutions

Complex Numbers

Q135 MCQ

For $(4 - 2i) + (1 + 5i)$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $5 + 3i$.

Correct option: C**Final answer** $5 + 3i$

Q136 MCQ

For $(2 + i)(2 - i)$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is 5.

Correct option: A**Final answer**

5

Worked Solutions

Complex Numbers

Q137 MCQ

For $(\sqrt{2} + 2i)^2$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $(\sqrt{2} + 2i)^2$.

Correct option: C**Final answer**

$$(\sqrt{2} + 2i)^2$$

Q138 MCQ

For $i^8 + i^{11} + i^{14}$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-i$.

Correct option: C**Final answer**

$$-i$$

Worked Solutions

Complex Numbers

Q139 MCQ

For $(3 - i)^3$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $(3 - i)^3$.

Correct option: D**Final answer**

$$(3 - i)^3$$

Q140 MCQ

For $(1 - 2i)(-2 + i)$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $5i$.

Correct option: D**Final answer**

$$5i$$

Worked Solutions

Complex Numbers

Q141 MCQ

For $i^5 + i^{10}$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-1 + i$.

Correct option: D**Final answer**

$$-1 + i$$

Q142 MCQ

For $(5 + 2i) - (3 - 4i)$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $2 + 6i$.

Correct option: D**Final answer**

$$2 + 6i$$

Worked Solutions

Complex Numbers

Q143 MCQ

For $(2 - 3i)^2$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $(2 - 3i)^2$.

Correct option: C

Final answer

$$(2 - 3i)^2$$

Q144 MCQ

For the expression State the complex conjugate of $3 - 2i$;, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $3 + 2i$.

Correct option: C

Final answer

$$3 + 2i$$

Worked Solutions

Complex Numbers

Q145 MCQ

For the expression State the complex conjugate of $-4 + \sqrt{3}i$; the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-4 - \sqrt{3}i$.

Correct option: A

Final answer

$$-4 - \sqrt{3}i$$

Q146 MCQ

For the expression State the complex conjugate of $7i$; the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-7i$.

Correct option: C

Final answer

$$-7i$$

Worked Solutions

Complex Numbers

Q147 MCQ

For the expression State the complex conjugate of $-\sqrt{5}$, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\sqrt{5}$.

Correct option: C

Final answer

$$-\sqrt{5}$$

Q148 MCQ

For the expression State the complex conjugate of $1/2 - 3i$, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{2} + 3i$.

Correct option: B

Final answer

$$\frac{1}{2} + 3i$$

Worked Solutions

Complex Numbers

Q149 MCQ

For $(2 + i)/(1 - i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{2} + \frac{3i}{2}$.

Correct option: A**Final answer**

$$\frac{1}{2} + \frac{3i}{2}$$

Q150 MCQ

For $(4 - 3i)/(2 + i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $1 - 2i$.

Correct option: A**Final answer**

$$1 - 2i$$

Worked Solutions

Complex Numbers

Q151 MCQ

For $(1 + 2i)/(3 - i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{10} + \frac{7i}{10}$.

Correct option: B**Final answer**

$$\frac{1}{10} + \frac{7i}{10}$$

Q152 MCQ

For $5i/(2 - i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $i(2 + i)$.

Correct option: C**Final answer**

$$i(2 + i)$$

Worked Solutions

Complex Numbers

Q153 MCQ

For $(3 - i)/(1 + 3i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-i$.

Correct option: D**Final answer** $-i$

Q154 MCQ

For $1/i$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-i$.

Correct option: D**Final answer** $-i$

Worked Solutions

Complex Numbers

Q155 MCQ

For $(2 + 3i)/(2 - 3i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{5}{13} + \frac{12i}{13}$.

Correct option: B**Final answer**

$$-\frac{5}{13} + \frac{12i}{13}$$

Q156 MCQ

For $(\sqrt{2} + i)/(\sqrt{2} - i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{\sqrt{2} + i}{\sqrt{2} - i}$.

Correct option: B**Final answer**

$$\frac{\sqrt{2} + i}{\sqrt{2} - i}$$

Worked Solutions

Complex Numbers

Q157 MCQ

For $(1 - 4i)/(3 + 2i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{5}{13} - \frac{14i}{13}$.

Correct option: D**Final answer**

$$-\frac{5}{13} - \frac{14i}{13}$$

Q158 MCQ

For $7i/(1 - i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{7}{2} + \frac{7i}{2}$.

Correct option: C**Final answer**

$$-\frac{7}{2} + \frac{7i}{2}$$

Worked Solutions

Complex Numbers

Q159 MCQ

For the expression State the complex conjugate of $-2 + 5i$., the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-2 - 5i$.

Correct option: B

Final answer

$$-2 - 5i$$

Q160 MCQ

For the expression State the complex conjugate of $6 - \sqrt{7}i$., the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $6 + \sqrt{7}i$.

Correct option: D

Final answer

$$6 + \sqrt{7}i$$

Worked Solutions

Complex Numbers

Q161 MCQ

For the expression State the complex conjugate of $-i\sqrt{2}$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\sqrt{2}i$.

Correct option: B**Final answer**

$$\sqrt{2}i$$

Q162 MCQ

For the expression State the complex conjugate of 9:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is 9.

Correct option: C**Final answer**

$$9$$

Worked Solutions

Complex Numbers

Q163 MCQ

For the expression State the complex conjugate of $-3/4 + 2i$., the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{3}{4} - 2i$.

Correct option: A**Final answer**

$$-\frac{3}{4} - 2i$$

Q164 MCQ

For $(5 + 2i)/(1 - 2i)$, select the value in the form $a+bi$:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{5} + \frac{12i}{5}$.

Correct option: D**Final answer**

$$\frac{1}{5} + \frac{12i}{5}$$

Worked Solutions

Complex Numbers

Q165 MCQ

For $(2 - i)/(4 + i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{7}{17} - \frac{6i}{17}$.

Correct option: A**Final answer**

$$\frac{7}{17} - \frac{6i}{17}$$

Q166 MCQ

For $3i/(2 + 3i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{9}{13} + \frac{6i}{13}$.

Correct option: A**Final answer**

$$\frac{9}{13} + \frac{6i}{13}$$

Worked Solutions

Complex Numbers

Q167 MCQ

For $(\sqrt{3} - i)/(\sqrt{3} + i)$, select the value in the form $a+bi$:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{\sqrt{3} - i}{\sqrt{3} + i}$.

Correct option: C

Final answer

$$\frac{\sqrt{3} - i}{\sqrt{3} + i}$$

Q168 MCQ

For $(1 + i)/(2i)$, select the value in the form $a+bi$:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{2} - \frac{i}{2}$.

Correct option: A

Final answer

$$\frac{1}{2} - \frac{i}{2}$$

Worked Solutions

Complex Numbers

Q169 MCQ

For the expression State the complex conjugate of $4 - 7i$., the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $4 + 7i$.

Correct option: D

Final answer

$$4 + 7i$$

Q170 MCQ

For the expression State the complex conjugate of $\sqrt{10} + 2i$., the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\sqrt{10} - 2i$.

Correct option: A

Final answer

$$\sqrt{10} - 2i$$

Worked Solutions

Complex Numbers

Q171 MCQ

For the expression State the complex conjugate of $-8i$, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $8i$.

Correct option: B

Final answer

$8i$

Q172 MCQ

For $(3 + 2i)/(3 - 2i)$, select the value in the form $a+bi$:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{5}{13} + \frac{12i}{13}$.

Correct option: B

Final answer

$\frac{5}{13} + \frac{12i}{13}$

Worked Solutions

Complex Numbers

Q173 MCQ

For $\sqrt{-5}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $i\sqrt{5}$.

Correct option: B**Final answer**

$$i\sqrt{5}$$

Q174 MCQ

For $\sqrt{-20}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $2i\sqrt{5}$.

Correct option: A**Final answer**

$$2i\sqrt{5}$$

Worked Solutions

Complex Numbers

Q175 MCQ

For $-\sqrt{-18}$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{-a} = i\sqrt{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is $-3i\sqrt{2}$.

Correct option: A**Final answer**

$$-3i\sqrt{2}$$

Q176 MCQ

For $\sqrt[4]{-8}$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt[4]{-a} = i\sqrt[4]{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is $\pm 2i\sqrt{2}$.

Correct option: C**Final answer**

$$\pm 2i\sqrt{2}$$

Worked Solutions

Complex Numbers

Q177 MCQ

For $\sqrt{-45}/\sqrt{-5}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is 3.

Correct option: A

Final answer

3

Q178 MCQ

For $\sqrt{-2}\sqrt{-8}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is -4 .

Correct option: B

Final answer -4

Worked Solutions

Complex Numbers

Q179 MCQ

For $\sqrt{12}/\sqrt{-3}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-2i$.

Correct option: C

Final answer $-2i$

Q180 MCQ

For $(2 + \sqrt{-3})^2$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $1 + 4i\sqrt{3}$.

Correct option: D

Final answer $1 + 4i\sqrt{3}$

Worked Solutions

Complex Numbers

Q181 MCQ

For $\sqrt{-72}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $6i\sqrt{2}$.

Correct option: D

Final answer

$$6i\sqrt{2}$$

Q182 MCQ

For $\sqrt{-50}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $\pm 5i\sqrt{2}$.

Correct option: D

Final answer

$$\pm 5i\sqrt{2}$$

Worked Solutions

Complex Numbers

Q183 MCQ

For $\sqrt{-3}\sqrt{-12}$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{-a} = i\sqrt{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is -6 .

Correct option: A

Final answer -6

Q184 MCQ

For $\sqrt{8}/\sqrt{-2}$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{-a} = i\sqrt{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is $-2i$.

Correct option: D

Final answer $-2i$

Worked Solutions

Complex Numbers

Q185 MCQ

For $\sqrt{-32} + \sqrt{-8}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $6i\sqrt{2}$.

Correct option: B

Final answer

$$6i\sqrt{2}$$

Q186 MCQ

For $\sqrt{-98} - \sqrt{-18}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $4i\sqrt{2}$.

Correct option: B

Final answer

$$4i\sqrt{2}$$

Worked Solutions

Complex Numbers

Q187 MCQ

For $(\sqrt{5} + \sqrt{-2})^2$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{a} + \sqrt{b})^2 = a + b + 2\sqrt{ab}$, then simplify the positive radical and the sign.
2. Therefore the correct value is $3 + 2i\sqrt{10}$.

Correct option: A**Final answer**

$$3 + 2i\sqrt{10}$$

Q188 MCQ

For $(3 + \sqrt{-5})(3 - \sqrt{-5})$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b$, then simplify the positive radical and the sign.
2. Therefore the correct value is 14.

Correct option: D**Final answer**

$$14$$

Worked Solutions

Complex Numbers

Q189 MCQ

For $\sqrt{-7}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $i\sqrt{7}$.

Correct option: C**Final answer**

$$i\sqrt{7}$$

Q190 MCQ

For $\sqrt{-28}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $2i\sqrt{7}$.

Correct option: A**Final answer**

$$2i\sqrt{7}$$

Worked Solutions

Complex Numbers

Q191 MCQ

For $\sqrt{-12}$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{-a} = i\sqrt{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is $\pm 2i\sqrt{3}$.

Correct option: A

Final answer

$$\pm 2i\sqrt{3}$$

Q192 MCQ

For $\sqrt{27}/\sqrt{-3}$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{-a} = i\sqrt{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is $-3i$.

Correct option: A

Final answer

$$-3i$$

Worked Solutions

Complex Numbers

Q193 MCQ

For $\sqrt{-2}\sqrt{-18}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is -6 .

Correct option: D

Final answer -6

Q194 MCQ

For $\sqrt{-48}/\sqrt{-3}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-4i$.

Correct option: C

Final answer $-4i$

Worked Solutions

Complex Numbers

Q195 MCQ

For $\sqrt{-8} + \sqrt{-18}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $5i\sqrt{2}$.

Correct option: C**Final answer**

$$5i\sqrt{2}$$

Q196 MCQ

For $(1 + \sqrt{-2})^2$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-1 + 2i\sqrt{2}$.

Correct option: A**Final answer**

$$-1 + 2i\sqrt{2}$$

Worked Solutions

Complex Numbers

Q197 MCQ

For $(4 + \sqrt{-3})(4 - \sqrt{-3})$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{-a} = i\sqrt{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is 19.

Correct option: B

Final answer

19

Q198 MCQ

For $\sqrt{-75}$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{-a} = i\sqrt{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is $5i\sqrt{3}$.

Correct option: D

Final answer $5i\sqrt{3}$

Worked Solutions

Complex Numbers

Q199 MCQ

Solve for the real values x and y : $(1 + 3i)(x + i) = x(y + i)$:**Solution**

1. Expand and equate real and imaginary parts.
2. The system gives $(x = -1/2)$ and $(y = 7)$.

Correct option: B**Final answer**

$$\left(-\frac{1}{2}, 7\right)$$

Q200 MCQ

Given x is real, solve: $(4x - 4i)(x^2 + xi) = 10x^2i^{32}$:**Solution**

1. Use $(i^{32} = 1)$ and $((x - i)(x + i) = x^2 + 1)$.
2. The real equation is $(4x(x^2 + 1) = 10x^2)$, giving $x = 0, 1/2$, or 2 .

Correct option: D**Final answer**

$$\left\{0, 2, \frac{1}{2}\right\}$$

Worked Solutions

Complex Numbers

Q201 MCQ

Simplify exactly: $(i(8 - 6i)/((1 + 2i)(3 - i)))^{2027}$:**Solution**

1. Rationalise the base using the conjugate of the denominator.
2. The exact base simplifies to $((7+i)/5)$. Keep the 2027th power unexpanded.

Correct option: B**Final answer**

$$((7 + i)/5)^{2027}$$

Q202 MCQ

Solve over the complex numbers: $(3x - 6)^2 = -8$:**Solution**

1. Take square roots: $(3x - 6 = \pm 2i\sqrt{2})$.
2. Divide by 3 to obtain $(x = 2 \pm \frac{2}{3}i\sqrt{2})$.

Correct option: B**Final answer**

$$x = 2 \pm (2/3)i\sqrt{2}$$

Worked Solutions

Complex Numbers

Quiz MCQ 1 [Concept Quiz · MCQ](#)

For $4-3i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $4 - 3i$ using the relevant family rule.
2. The correct value is $(4, -3)$.

Correct option: A

Final answer

$(4, -3)$

Quiz MCQ 2 [Concept Quiz · MCQ](#)

For $(1+2i)(2-i)$, select the simplified value:

Solution

1. Simplify $(1 + 2i)(2 - i)$ using the relevant family rule.
2. The correct value is $4 + 3i$.

Correct option: A

Final answer

$4 + 3i$

Worked Solutions

Complex Numbers

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)

For $(3+i)/(1-i)$, select the value in the form $a+bi$:

Solution

1. Simplify $(3+i)/(1-i)$ using the relevant family rule.
2. The correct value is $1+2i$.

Correct option: D

Final answer

$$1 + 2i$$

Quiz MCQ 4 [Concept Quiz](#) · [MCQ](#)

For $\sqrt{-27}$, select the expression in terms of i :

Solution

1. Simplify $\sqrt{-27}$ using the relevant family rule.
2. The correct value is $3i\sqrt{3}$.

Correct option: A

Final answer

$$3i\sqrt{3}$$

Worked Solutions

Complex Numbers

Quiz WRITTEN 5 Concept Quiz · Written

State the real and imaginary parts of $\frac{5-\sqrt{5}i}{2}$.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: $(5/2, -\sqrt{5}/2)$

Final answer

$(5/2, -\sqrt{5}/2)$

Quiz WRITTEN 6 Concept Quiz · Written

Solve for real x, y : $(2 + i)x + (1 - i)y = 3 + i$.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: $x = 4/3, y = 1/3$

Final answer

$x = 4/3, y = 1/3$

Worked Solutions

Complex Numbers

Quiz WRITTEN 7 Concept Quiz · Written

Calculate $i^{18} + i^7 - i^3$ in standard form.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: -1

Final answer

 -1

Quiz WRITTEN 8 Concept Quiz · Written

Solve over the complex numbers: $(x - 3)^2 = -12$.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: $x = 3 \pm 2i\sqrt{3}$

Final answer

 $x = 3 \pm 2i\sqrt{3}$

Answer key

Use the question number shown on each page.

Q001 $(3, -2)$	Q027 $3 - 4i$	Q053 2	Q079 $3i\sqrt{3}$
Q002 $(4/3, -\sqrt{5}/3)$	Q028 -1	Q054 $8i$	Q080 $-2i\sqrt{3}$
Q003 $(-8, 0)$	Q029 $1 + 8i$	Q055 $1 - i$	Q081 $\pm 2i\sqrt{3}$
Q004 $(0, 2)$	Q030 $8 - i$	Q056 $-1 - i$	Q082 $-3\sqrt{2}$
Q005 $x = -1, y = 4$	Q031 0	Q057 $3 - 4i$	Q083 $-3i$
Q006 $(-3, 5)$	Q032 $4 - i$	Q058 $9 - 8i$	Q084 $7 + 6\sqrt{2}i$
Q007 $(-1/2, -\sqrt{3}/2)$	Q033 $1 - 3i$	Q059 $8 - i\sqrt{3}$	Q085 $i\sqrt{2}$
Q008 $(1, 0)$	Q034 -2	Q060 5	Q086 $3i$
Q009 $(0, 3)$	Q035 $-1 + 4\sqrt{2}i$	Q061 -15	Q087 $3i\sqrt{5}$
Q010 $x = 6, y = -2$	Q036 $7 + 11i$	Q062 $\sqrt{7}$	Q088 $-2i\sqrt{6}$
Q011 $x = -2, y = 3$	Q037 $1/15 - 13/14i$	Q063 $-i$	Q089 $-6i$
Q012 $(2, 5)$	Q038 $-3 - 4i$	Q064 $50i$	Q090 $\pm i\sqrt{2}$
Q013 $(-25, 13)$	Q039 -3	Q065 $-i\sqrt{6}$	Q091 $\pm 3i\sqrt{2}$
Q014 $(2/3, -\sqrt{5}/3)$	Q040 4	Q066 i	Q092 $\pm 4i$
Q015 $(7/2, -\sqrt{2}/2)$	Q041 i	Q067 $7/5 - 1/5i$	Q093 $-2\sqrt{3}$
Q016 $(-4, 0)$	Q042 -1	Q068 $2/3 + \sqrt{5}/3i$	Q094 $-14\sqrt{5}$
Q017 $(1/4, 0)$	Q043 $-i$	Q069 $2 + i$	Q095 $-2i$
Q018 $(0, \sqrt{3})$	Q044 $x = 0$ or $x = 1$	Q070 $3/5 + 4/5i$	Q096 $-(3/2)i$
Q019 $(0, -1)$	Q045 $2 - 3i$	Q071 $9/13 - 7/13i$	Q097 $13i\sqrt{2}$
Q020 $x = 3, y = 2$	Q046 $7 + i\sqrt{11}$	Q072 $-i$	Q098 $6i\sqrt{2}$
Q021 $x = 3, y = -3$	Q047 6	Q073 $1/3 - 2/3i$	Q099 $-2 + 2i\sqrt{15}$
Q022 $x = 2, y = -2$	Q048 $5i$	Q074 $((-2 - 144i)/85)^{2027}$	Q100 7
Q023 $x = 2, y = 1$	Q049 $3/5 + 4/5i$	Q075 $\pm i\sqrt{6}$	Q101 $x = 4 \pm (\sqrt{10}/2)i$
Q024 $x = 1/2, y = 3/2$	Q050 $-1/2 - 3/2i$	Q076 $-6\sqrt{2}$	Q102 C
Q025 $x = -1, y = 3$	Q051 $1 + 4i$	Q077 $-(2/3)i$	Q103 D
Q026 $1 - 7i$	Q052 $\sqrt{2} - i\sqrt{10}$	Q078 $i\sqrt{7}$	Q104 D

Q105 B	Q132 B	Q159 B	Q186 B
Q106 D	Q133 B	Q160 D	Q187 A
Q107 A	Q134 B	Q161 B	Q188 D
Q108 B	Q135 C	Q162 C	Q189 C
Q109 C	Q136 A	Q163 A	Q190 A
Q110 B	Q137 C	Q164 D	Q191 A
Q111 D	Q138 C	Q165 A	Q192 A
Q112 D	Q139 D	Q166 A	Q193 D
Q113 C	Q140 D	Q167 C	Q194 C
Q114 D	Q141 D	Q168 A	Q195 C
Q115 B	Q142 D	Q169 D	Q196 A
Q116 B	Q143 C	Q170 A	Q197 B
Q117 C	Q144 C	Q171 B	Q198 D
Q118 D	Q145 A	Q172 B	Q199 B
Q119 A	Q146 C	Q173 B	Q200 D
Q120 B	Q147 C	Q174 A	Q201 B
Q121 B	Q148 B	Q175 A	Q202 B
Q122 C	Q149 A	Q176 C	Quiz MCQ 1 A
Q123 D	Q150 A	Q177 A	Quiz MCQ 2 A
Q124 D	Q151 B	Q178 B	Quiz MCQ 3 D
Q125 C	Q152 C	Q179 C	Quiz MCQ 4 A
Q126 C	Q153 D	Q180 D	Quiz WRITTEN 5 $(5/2, -\sqrt{5}/2)$
Q127 C	Q154 D	Q181 D	Quiz WRITTEN 6 $x = 4/3, y = 1/3$
Q128 D	Q155 B	Q182 D	Quiz WRITTEN 7 -1
Q129 A	Q156 B	Q183 A	Quiz WRITTEN 8 $x = 3 \pm 2i\sqrt{3}$
Q130 C	Q157 D	Q184 D	
Q131 A	Q158 C	Q185 B	

Concept 2 | Quadratic Equations

English Student Edition • Focused formulas and guided practice

Formula opener

Roots

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum/product

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}$$

Discriminant

$$\Delta = b^2 - 4ac$$

Worked Solutions

Quadratic Equations

Q001 Written

Solve $x^2 + 8 = 0$.**Solution**

1. $x^2 = -8$
2. $x = \pm\sqrt{-8} = \pm i\sqrt{8} = \pm 2i\sqrt{2}$
3. Therefore the final answer is $x = \pm 2i\sqrt{2}$.

Final answer

$$x = \pm 2i\sqrt{2}$$

Q002 Written

Solve $x^2 + 4 = 0$.**Solution**

1. $x^2 = -4$
2. $x = \pm 2i$
3. Therefore the final answer is $x = \pm 2i$.

Final answer

$$x = \pm 2i$$

Worked Solutions

Quadratic Equations

Q003 Written

Solve $x^2 + 49 = 0$.

Solution

1. $x^2 = -49$
2. Therefore the final answer is $x = \pm 7i$.

Final answer

$$x = \pm 7i$$

Q004 Written

Solve $9x^2 - 4 = 0$.

Solution

1. $(3x - 2)(3x + 2) = 0$
2. Therefore the final answer is $x = \pm \frac{2}{3}$.

Final answer

$$x = \pm \frac{2}{3}$$

Worked Solutions

Quadratic Equations

Q005 Written

Solve $x^2 - 12 = 0$.**Solution**

1. $x = \pm\sqrt{12}$
2. Therefore the final answer is $x = \pm 2\sqrt{3}$.

Final answer

$$x = \pm 2\sqrt{3}$$

Q006 Written

Solve $4x^2 + 3 = 0$.**Solution**

1. $x^2 = -3/4$
2. Therefore the final answer is $x = \pm \frac{i\sqrt{3}}{2}$.

Final answer

$$x = \pm \frac{i\sqrt{3}}{2}$$

Worked Solutions

Quadratic Equations

Q007 MCQ

Solve $x^2 + 8 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \pm 2i\sqrt{2}$.

Correct option: A**Final answer**

$$x = \pm 2i\sqrt{2}$$

Q008 MCQ

Solve $x^2 + 4 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \pm 2i$.

Correct option: A**Final answer**

$$x = \pm 2i$$

Worked Solutions

Quadratic Equations

Q009 MCQ

Solve $x^2 + 49 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \pm 7i$.

Correct option: C**Final answer**

$$x = \pm 7i$$

Q010 MCQ

Solve $9x^2 - 4 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \pm \frac{2}{3}$.

Correct option: C**Final answer**

$$x = \pm \frac{2}{3}$$

Worked Solutions

Quadratic Equations

Q011 MCQ

Solve $x^2 - 12 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \pm 2\sqrt{3}$.

Correct option: B**Final answer**

$$x = \pm 2\sqrt{3}$$

Q012 MCQ

Solve $4x^2 + 3 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \pm \frac{i\sqrt{3}}{2}$.

Correct option: B**Final answer**

$$x = \pm \frac{i\sqrt{3}}{2}$$

Worked Solutions

Quadratic Equations

Q013 Written

Solve $(x - 3)^2 = -8$.**Solution**

1. $x - 3 = \pm i\sqrt{8} = \pm 2i\sqrt{2}$
2. Therefore the final answer is $x = 3 \pm 2i\sqrt{2}$.

Final answer

$$x = 3 \pm 2i\sqrt{2}$$

Q014 Written

Solve $(x - 2)^2 = 5$.**Solution**

1. $x - 2 = \pm\sqrt{5}$
2. Therefore the final answer is $x = 2 \pm \sqrt{5}$.

Final answer

$$x = 2 \pm \sqrt{5}$$

Worked Solutions

Quadratic Equations

Q015 Written

Solve $(x + 6)^2 = -6$.**Solution**

1. $x + 6 = \pm i\sqrt{6}$
2. Therefore the final answer is $x = -6 \pm i\sqrt{6}$.

Final answer

$$x = -6 \pm i\sqrt{6}$$

Q016 Written

Solve $(x + 4)^2 = -18$.**Solution**

1. $x + 4 = \pm i\sqrt{18} = \pm 3i\sqrt{2}$
2. Therefore the final answer is $x = -4 \pm 3i\sqrt{2}$.

Final answer

$$x = -4 \pm 3i\sqrt{2}$$

Worked Solutions

Quadratic Equations

Q017 **Written****Solve** $(2x - 3)^2 = 7$.**Solution**

1. $2x - 3 = \pm\sqrt{7}$

2. Therefore the final answer is $x = \frac{3 \pm \sqrt{7}}{2}$.

Final answer

$$x = \frac{3 \pm \sqrt{7}}{2}$$

Q018 **MCQ****Solve** $(x - 3)^2 = -8$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = 3 \pm 2i\sqrt{2}$.**Correct option:** A**Final answer**

$$x = 3 \pm 2i\sqrt{2}$$

Worked Solutions

Quadratic Equations

Q019 MCQ

Solve $(x - 2)^2 = 5$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = 2 \pm \sqrt{5}$.

Correct option: B**Final answer**

$$x = 2 \pm \sqrt{5}$$

Q020 MCQ

Solve $(x + 6)^2 = -6$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = -6 \pm i\sqrt{6}$.

Correct option: C**Final answer**

$$x = -6 \pm i\sqrt{6}$$

Worked Solutions

Quadratic Equations

Q021 MCQ

Solve $(x + 4)^2 = -18$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = -4 \pm 3i\sqrt{2}$.

Correct option: C**Final answer**

$$x = -4 \pm 3i\sqrt{2}$$

Q022 MCQ

Solve $(2x - 3)^2 = 7$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \frac{3 \pm \sqrt{7}}{2}$.

Correct option: D**Final answer**

$$x = \frac{3 \pm \sqrt{7}}{2}$$

Worked Solutions

Quadratic Equations

Q023 Written

Solve $2x^2 - 7x - 9 = 0$.**Solution**

1. $(x + 1)(2x - 9) = 0$
2. $x = -1$ or $x = 9/2$
3. Therefore the final answer is $x = -1, \frac{9}{2}$.

Final answer

$$x = -1, \frac{9}{2}$$

Q024 Written

Solve $2x^2 + 5x - 3 = 0$.**Solution**

1. $(2x - 1)(x + 3) = 0$
2. Therefore the final answer is $x = \frac{1}{2}, -3$.

Final answer

$$x = \frac{1}{2}, -3$$

Worked Solutions

Quadratic Equations

Q025 Written

Solve $2x^2 + 7x = 0$.**Solution**

1. $x(2x + 7) = 0$
2. Therefore the final answer is $x = 0, -\frac{7}{2}$.

Final answer

$$x = 0, -\frac{7}{2}$$

Q026 Written

Solve $2x^2 + 5x - 12 = 0$.**Solution**

1. $(2x - 3)(x + 4) = 0$
2. Therefore the final answer is $x = \frac{3}{2}, -4$.

Final answer

$$x = \frac{3}{2}, -4$$

Worked Solutions

Quadratic Equations

Q027 **Written****Solve** $x^2 - 8x + 15 = 0$.**Solution**

1. $(x - 3)(x - 5) = 0$
2. Therefore the final answer is $x = 3, 5$.

Final answer $x = 3, 5$ Q028 **MCQ****Solve** $2x^2 - 7x - 9 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = -1, \frac{9}{2}$.

Correct option: A**Final answer** $x = -1, \frac{9}{2}$

Worked Solutions

Quadratic Equations

Q029 MCQ

Solve $2x^2 + 5x - 3 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \frac{1}{2}, -3$.

Correct option: D**Final answer**

$$x = \frac{1}{2}, -3$$

Q030 MCQ

Solve $2x^2 + 7x = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = 0, -\frac{7}{2}$.

Correct option: D**Final answer**

$$x = 0, -\frac{7}{2}$$

Worked Solutions

Quadratic Equations

Q031 MCQ

Solve $2x^2 + 5x - 12 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \frac{3}{2}, -4$.

Correct option: A**Final answer**

$$x = \frac{3}{2}, -4$$

Q032 MCQ

Solve $x^2 - 8x + 15 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = 3, 5$.

Correct option: A**Final answer**

$$x = 3, 5$$

Worked Solutions

Quadratic Equations

Q033 Written

Solve $3x^2 - 4x - 1 = 0$.**Solution**

1. $D = 16 + 12 = 28$
2. $x = (4 \pm 2\sqrt{7})/6 = (2 \pm \sqrt{7})/3$
3. Therefore the final answer is $x = \frac{2 \pm \sqrt{7}}{3}$.

Final answer

$$x = \frac{2 \pm \sqrt{7}}{3}$$

Q034 MCQ

Solve $3x^2 - 4x - 1 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \frac{2 \pm \sqrt{7}}{3}$.

Correct option: B**Final answer**

$$x = \frac{2 \pm \sqrt{7}}{3}$$

Worked Solutions

Quadratic Equations

Q035 Written

Solve $2x^2 + 2\sqrt{6}x + 11 = 0$.**Solution**

1. $D = (2\sqrt{6})^2 - 4(2)(11) = -64$

2. $x = (-2\sqrt{6} \pm \sqrt{(-64)})/4 = (-\sqrt{6} \pm 4i)/2$

3. Therefore the final answer is $x = \frac{-\sqrt{6} \pm 4i}{2}$.

Final answer

$$x = \frac{-\sqrt{6} \pm 4i}{2}$$

Q036 Written

Solve $2x^2 + 2\sqrt{3}x + 5 = 0$.**Solution**

1. $D = 12 - 40 = -28$

2. $x = (-2\sqrt{3} \pm i\sqrt{28})/4 = (-\sqrt{3} \pm i\sqrt{7})/2$

3. Therefore the final answer is $x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$.

Final answer

$$x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$$

Worked Solutions

Quadratic Equations

Q037 Written

Solve $x^2 + x + 1 = 0$.**Solution**

1. $D = 1 - 4 = -3$
2. $x = (-1 \pm i\sqrt{3})/2$
3. Therefore the final answer is $x = \frac{-1 \pm i\sqrt{3}}{2}$.

Final answer

$$x = \frac{-1 \pm i\sqrt{3}}{2}$$

Q038 Written

Solve $3x^2 - 5x + 3 = 0$.**Solution**

1. $D = 25 - 36 = -11$
2. $x = (5 \pm i\sqrt{11})/6$
3. Therefore the final answer is $x = \frac{5 \pm i\sqrt{11}}{6}$.

Final answer

$$x = \frac{5 \pm i\sqrt{11}}{6}$$

Worked Solutions

Quadratic Equations

Q039 Written

Solve $x^2 - 2\sqrt{3}x + 5 = 0$.**Solution**

1. $D = 12 - 20 = -8$
2. $x = (2\sqrt{3} \pm i\sqrt{8})/2 = \sqrt{3} \pm i\sqrt{2}$
3. Therefore the final answer is $x = \sqrt{3} \pm i\sqrt{2}$.

Final answer

$$x = \sqrt{3} \pm i\sqrt{2}$$

Q040 Written

Solve $3x^2 + 2\sqrt{2}x + 1 = 0$.**Solution**

1. $D = 8 - 12 = -4$
2. $x = (-2\sqrt{2} \pm 2i)/6 = (-\sqrt{2} \pm i)/3$
3. Therefore the final answer is $x = \frac{-\sqrt{2} \pm i}{3}$.

Final answer

$$x = \frac{-\sqrt{2} \pm i}{3}$$

Worked Solutions

Quadratic Equations

Q041 Written

Solve $x^2 - x + 2 = 0$.**Solution**

1. $D = 1 - 8 = -7$
2. $x = (1 \pm i\sqrt{7})/2$
3. Therefore the final answer is $x = \frac{1 \pm i\sqrt{7}}{2}$.

Final answer

$$x = \frac{1 \pm i\sqrt{7}}{2}$$

Q042 Written

Solve $3x^2 - \sqrt{5}x + 1 = 0$.**Solution**

1. $D = 5 - 12 = -7$
2. $x = (\sqrt{5} \pm i\sqrt{7})/6$
3. Therefore the final answer is $x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$.

Final answer

$$x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$$

Worked Solutions

Quadratic Equations

Q043 Written

Solve $6x^2 - 12x + 15 = 0$.**Solution**

1. $D = 144 - 360 = -216$
2. $x = (12 \pm i6\sqrt{6})/12 = 1 \pm i\sqrt{6}/2$
3. Therefore the final answer is $x = 1 \pm \frac{i\sqrt{6}}{2}$.

Final answer

$$x = 1 \pm \frac{i\sqrt{6}}{2}$$

Q044 MCQ

Solve $2x^2 + 2\sqrt{6}x + 11 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \frac{-\sqrt{6} \pm 4i}{2}$.

Correct option: B**Final answer**

$$x = \frac{-\sqrt{6} \pm 4i}{2}$$

Worked Solutions

Quadratic Equations

Q045 MCQ

Solve $2x^2 + 2\sqrt{3}x + 5 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$.**Correct option:** C**Final answer**

$$x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$$

Q046 MCQ

Solve $x^2 + x + 1 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = \frac{-1 \pm i\sqrt{3}}{2}$.**Correct option:** D**Final answer**

$$x = \frac{-1 \pm i\sqrt{3}}{2}$$

Worked Solutions

Quadratic Equations

Q047 MCQ

Solve $3x^2 - 5x + 3 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = \frac{5 \pm i\sqrt{11}}{6}$.**Correct option:** B**Final answer**

$$x = \frac{5 \pm i\sqrt{11}}{6}$$

Q048 MCQ

Solve $x^2 - 2\sqrt{3}x + 5 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = \sqrt{3} \pm i\sqrt{2}$.**Correct option:** D**Final answer**

$$x = \sqrt{3} \pm i\sqrt{2}$$

Worked Solutions

Quadratic Equations

Q049 MCQ

Solve $3x^2 + 2\sqrt{2}x + 1 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = \frac{-\sqrt{2} \pm i}{3}$.**Correct option:** B**Final answer**

$$x = \frac{-\sqrt{2} \pm i}{3}$$

Q050 MCQ

Solve $x^2 - x + 2 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = \frac{1 \pm i\sqrt{7}}{2}$.**Correct option:** C**Final answer**

$$x = \frac{1 \pm i\sqrt{7}}{2}$$

Worked Solutions

Quadratic Equations

Q051 MCQ

Solve $3x^2 - \sqrt{5}x + 1 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$.**Correct option:** C**Final answer**

$$x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$$

Q052 MCQ

Solve $6x^2 - 12x + 15 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = 1 \pm \frac{i\sqrt{6}}{2}$.**Correct option:** B**Final answer**

$$x = 1 \pm \frac{i\sqrt{6}}{2}$$

Worked Solutions

Quadratic Equations

Q053 Written

Determine the nature of the solutions of $3x^2 + 4x + 7 = 0$.**Solution**

1. $D = 4^2 - 4(3)(7) = -68 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Q054 Written

Find the range of m for which $x^2 - mx + m^2 - 3m - 9 = 0$ has real solutions.**Solution**

1. $D = (-m)^2 - 4(m^2 - 3m - 9) = -3m^2 + 12m + 36$
2. $D \geq 0 \rightarrow m^2 - 4m - 12 \leq 0 \rightarrow (m + 2)(m - 6) \leq 0$
3. Therefore the final answer is $-2 \leq m \leq 6$.

Final answer $-2 \leq m \leq 6$

Worked Solutions

Quadratic Equations

Q055 Written

For $x^2 + kx + 3 - k = 0$, determine the nature of the solutions as k varies.

Solution

1. $D = k^2 - 4(3 - k) = (k - 2)(k + 6)$
2. Use $D > 0, D = 0, D < 0$ intervals
3. Therefore the final answer is $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary.

Final answer

$k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary

Q056 Written

For $3x^2 - 5x + 4 = 0$, determine the nature of the solutions.

Solution

1. $D = 25 - 48 = -23 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Worked Solutions

Quadratic Equations

Q057 **Written****For $4x^2 - 7x - 2 = 0$, determine the nature of the solutions.****Solution**

1. $D = 49 + 32 = 81 > 0$
2. Therefore the final answer is Two distinct real solutions.

Final answer

Two distinct real solutions

Q058 **Written****For $25x^2 - 10x + 1 = 0$, determine the nature of the solutions.****Solution**

1. $D = 100 - 100 = 0$
2. Therefore the final answer is One repeated real solution.

Final answer

One repeated real solution

Worked Solutions

Quadratic Equations

Q059 Written

Determine the nature of the solutions of $x^2 - 5x + 3 = 0$.**Solution**

1. $D = 25 - 12 = 13 > 0$
2. Therefore the final answer is Two distinct real solutions.

Final answer

Two distinct real solutions

Q060 Written

Determine the nature of the solutions of $9x^2 - 12x + 4 = 0$.**Solution**

1. $D = 144 - 144 = 0$
2. Therefore the final answer is One repeated real solution.

Final answer

One repeated real solution

Worked Solutions

Quadratic Equations

Q061 Written

Determine the nature of the solutions of $4x^2 - 8x + 5 = 0$.**Solution**

1. $D = 64 - 80 = -16 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Q062 Written

Determine the nature of the solutions of $4x^2 - 4x + 1 = 0$.**Solution**

1. $D = 16 - 16 = 0$
2. Therefore the final answer is One repeated real solution.

Final answer

One repeated real solution

Worked Solutions

Quadratic Equations

Q063 Written

Determine the nature of the solutions of $x^2 - 3x + 5 = 0$.

Solution

1. $D = 9 - 20 = -11 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Q064 Written

Determine the nature of the solutions of $x^2 - x - 2 = 0$.

Solution

1. $D = 1 + 8 = 9 > 0$
2. Therefore the final answer is Two distinct real solutions.

Final answer

Two distinct real solutions

Worked Solutions

Quadratic Equations

Q065 MCQ

Determine the nature of the solutions of $3x^2 + 4x + 7 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: A**Final answer**

Two distinct imaginary (non-real complex) solutions

Worked Solutions

Quadratic Equations

Q066 MCQ

Find the range of m for which $x^2 - mx + m^2 - 3m - 9 = 0$ has real solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-2 \leq m \leq 6$.

Correct option: D

Final answer

$$-2 \leq m \leq 6$$

Worked Solutions

Quadratic Equations

Q067 MCQ

For $x^2 + kx + 3 - k = 0$, determine the nature of the solutions as k varies. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary.

Correct option: C

Final answer

$k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary

Worked Solutions

Quadratic Equations

Q068 MCQ

For $3x^2 - 5x + 4 = 0$, determine the nature of the solutions. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: C

Final answer

Two distinct imaginary (non-real complex) solutions

Worked Solutions

Quadratic Equations

Q069 MCQ

For $4x^2 - 7x - 2 = 0$, determine the nature of the solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct real solutions.

Correct option: A

Final answer

Two distinct real solutions

Worked Solutions

Quadratic Equations

Q070 MCQ

For $25x^2 - 10x + 1 = 0$, determine the nature of the solutions. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is One repeated real solution.

Correct option: A

Final answer

One repeated real solution

Worked Solutions

Quadratic Equations

Q071 MCQ

Determine the nature of the solutions of $x^2 - 5x + 3 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct real solutions.

Correct option: C**Final answer**

Two distinct real solutions

Worked Solutions

Quadratic Equations

Q072 MCQ

Determine the nature of the solutions of $9x^2 - 12x + 4 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is One repeated real solution.

Correct option: C**Final answer**

One repeated real solution

Worked Solutions

Quadratic Equations

Q073 MCQ

Determine the nature of the solutions of $4x^2 - 8x + 5 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: B**Final answer**

Two distinct imaginary (non-real complex) solutions

Worked Solutions

Quadratic Equations

Q074 MCQ

Determine the nature of the solutions of $4x^2 - 4x + 1 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is One repeated real solution.

Correct option: A**Final answer**

One repeated real solution

Worked Solutions

Quadratic Equations

Q075 MCQ

Determine the nature of the solutions of $x^2 - 3x + 5 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: A**Final answer**

Two distinct imaginary (non-real complex) solutions

Worked Solutions

Quadratic Equations

Q076 MCQ

Determine the nature of the solutions of $x^2 - x - 2 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct real solutions.

Correct option: A

Final answer

Two distinct real solutions

Worked Solutions

Quadratic Equations

Q077 Written

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary solutions.

Solution

1. $D = 4m^2 - 4(m + 2)$
2. $D < 0 \rightarrow m^2 - m - 2 < 0 \rightarrow (m - 2)(m + 1) < 0$
3. Therefore the final answer is $-1 < m < 2$.

Final answer

$$-1 < m < 2$$

Q078 Written

For $x^2 - ax + 2a - 3 = 0$, determine the nature of the solutions as a varies.

Solution

1. $D = a^2 - 4(2a - 3) = (a - 2)(a - 6)$
2. Classify by sign of D
3. Therefore the final answer is $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary.

Final answer $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

Worked Solutions

Quadratic Equations

Q079 Written

Find the values of a for which $x^2 - (a + 1)x + 4 = 0$ has a repeated solution.

Solution

1. $D = (a + 1)^2 - 16 = 0$
2. $a + 1 = \pm 4$
3. Therefore the final answer is $a = 3$ or $a = -5$.

Final answer $a = 3$ or $a = -5$

Q080 Written

Find the range of m for which $x^2 + (m + 6)x - 2m = 0$ has two distinct imaginary solutions.

Solution

1. $D = (m + 6)^2 + 8m = m^2 + 20m + 36 = (m + 2)(m + 18)$
2. $D < 0$
3. Therefore the final answer is $-18 < m < -2$.

Final answer $-18 < m < -2$

Worked Solutions

Quadratic Equations

Q081 Written

Find the range of m for which $x^2 - 2mx + 2m^2 - 3m = 0$ has real solutions.

Solution

1. $D = 4m^2 - 4(2m^2 - 3m) = 4m(3 - m)$
2. $D \geq 0$
3. Therefore the final answer is $0 \leq m \leq 3$.

Final answer

$$0 \leq m \leq 3$$

Q082 MCQ

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-1 < m < 2$.

Correct option: C

Final answer

$$-1 < m < 2$$

Worked Solutions

Quadratic Equations

Q083 MCQ

For $x^2 - ax + 2a - 3 = 0$, determine the nature of the solutions as a varies. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary.

Correct option: C

Final answer

$a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

Worked Solutions

Quadratic Equations

Q084 MCQ

Find the values of a for which $x^2 - (a + 1)x + 4 = 0$ has a repeated solution Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $a = 3$ or $a = -5$.

Correct option: A**Final answer** $a = 3$ or $a = -5$

Q085 MCQ

Find the range of m for which $x^2 + (m + 6)x - 2m = 0$ has two distinct imaginary solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-18 < m < -2$.

Correct option: D**Final answer** $-18 < m < -2$

Worked Solutions

Quadratic Equations

Q086 MCQ

Find the range of m for which $x^2 - 2mx + 2m^2 - 3m = 0$ has real solutions. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $0 \leq m \leq 3$.

Correct option: C

Final answer

$$0 \leq m \leq 3$$

Q087 Written

Find the sum and product of the two roots of $3x^2 - 6x - 5 = 0$.

Solution

1. Use $S = -b/a = 6/3 = 2$ and $P = c/a = -5/3$
2. Therefore the final answer is $\alpha + \beta = 2$, $\alpha\beta = -\frac{5}{3}$.

Final answer

$$\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3}$$

Worked Solutions

Quadratic Equations

Q088 Written

Find the sum and product of the roots of $4x^2 + 2x - 7 = 0$.**Solution**

1. $S = -2/4 = -1/2$; $P = -7/4$
2. Therefore the final answer is $\alpha + \beta = -\frac{1}{2}$, $\alpha\beta = -\frac{7}{4}$.

Final answer

$$\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$$

Q089 Written

Find the sum and product of the roots of $x^2 + x + 2 = 0$.**Solution**

1. $S = -1$; $P = 2$
2. Therefore the final answer is $S = -1$, $P = 2$.

Final answer

$$S = -1, \quad P = 2$$

Worked Solutions

Quadratic Equations

Q090 Written

Find the sum and product of the roots of $3x^2 - 7x - 6 = 0$.**Solution**

1. $S = 7/3; P = -2$
2. Therefore the final answer is $S = \frac{7}{3}, P = -2$.

Final answer

$$S = \frac{7}{3}, P = -2$$

Q091 MCQ

Find the sum and product of the two roots of $3x^2 - 6x - 5 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $\alpha + \beta = 2, \alpha\beta = -\frac{5}{3}$.

Correct option: C**Final answer**

$$\alpha + \beta = 2, \alpha\beta = -\frac{5}{3}$$

Worked Solutions

Quadratic Equations

Q092 MCQ

Find the sum and product of the roots of $4x^2 + 2x - 7 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $\alpha + \beta = -\frac{1}{2}$, $\alpha\beta = -\frac{7}{4}$.

Correct option: A**Final answer**

$$\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$$

Q093 MCQ

Find the sum and product of the roots of $x^2 + x + 2 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $S = -1$, $P = 2$.

Correct option: B**Final answer**

$$S = -1, \quad P = 2$$

Worked Solutions

Quadratic Equations

Q094 MCQ

Find the sum and product of the roots of $3x^2 - 7x - 6 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $S = \frac{7}{3}$, $P = -2$.

Correct option: A

Final answer

$$S = \frac{7}{3}, \quad P = -2$$

Q095 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^2 + \beta^2$.

Solution

1. $S = 3, P = -3/2$
2. $S^2 - 2P = 9 + 3 = 12$
3. Therefore the final answer is 12.

Final answer

12

Worked Solutions

Quadratic Equations

Q096 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 3, P = -3/2$
2. $S^3 - 3PS = 27 + 27/2 = 81/2$
3. Therefore the final answer is $\frac{81}{2}$.

Final answer

$$\frac{81}{2}$$

Q097 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^2 + \beta^2$.

Solution

1. $S = 2, P = 3/2$
2. $S^2 - 2P = 4 - 3 = 1$
3. Therefore the final answer is 1.

Final answer

$$1$$

Worked Solutions

Quadratic Equations

Q098 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 2, P = 3/2$
2. $S^3 - 3PS = 8 - 9 = -1$
3. Therefore the final answer is -1 .

Final answer -1

Q099 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^2 + \beta^2$.

Solution

1. $S = 2, P = 3$
2. $S^2 - 2P = 4 - 6 = -2$
3. Therefore the final answer is -2 .

Final answer -2

Worked Solutions

Quadratic Equations

Q100 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 2, P = 3$
2. $S^3 - 3PS = 8 - 18 = -10$
3. Therefore the final answer is -10 .

Final answer

-10

Q101 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^2 + \beta^2$.

Solution

1. $S = -2, P = -1/3$
2. $S^2 - 2P = 4 + 2/3 = 14/3$
3. Therefore the final answer is $\frac{14}{3}$.

Final answer

$\frac{14}{3}$

Worked Solutions

Quadratic Equations

Q102 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = -2, P = -1/3$
2. $S^3 - 3PS = -8 - 2 = -10$
3. Therefore the final answer is -10 .

Final answer

-10

Q103 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 12.

Correct option: A

Final answer

12

Worked Solutions

Quadratic Equations

Q104 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{81}{2}$.

Correct option: B**Final answer**

$$\frac{81}{2}$$

Q105 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 1.

Correct option: A**Final answer**

$$1$$

Worked Solutions

Quadratic Equations

Q106 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is -1 .

Correct option: B**Final answer** -1

Q107 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is -2 .

Correct option: D**Final answer** -2

Worked Solutions

Quadratic Equations

Q108 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is -10 .

Correct option: B**Final answer** -10

Q109 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{14}{3}$.

Correct option: B**Final answer** $\frac{14}{3}$

Worked Solutions

Quadratic Equations

Q110 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^3 + \beta^3$. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is -10 .

Correct option: D**Final answer** -10

Q111 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\frac{2}{\alpha} + \frac{2}{\beta}$.

Solution

1. $2(\alpha + \beta)/(\alpha\beta) = 2(3)/(-3/2) = -4$
2. Therefore the final answer is -4 .

Final answer -4

Worked Solutions

Quadratic Equations

Q112 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\frac{\beta}{\alpha} + \frac{\alpha}{\beta}$.

Solution

1. $(\alpha^2 + \beta^2)/(\alpha\beta) = 1/(3/2) = 2/3$
2. Therefore the final answer is $\frac{2}{3}$.

Final answer

$$\frac{2}{3}$$

Q113 Written

Find the sum and product of the roots of $5x^2 - 3 = 0$.

Solution

1. $S = 0; P = -3/5$
2. Therefore the final answer is $S = 0, P = -\frac{3}{5}$.

Final answer

$$S = 0, P = -\frac{3}{5}$$

Worked Solutions

Quadratic Equations

Q114 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$.

Solution

1. $S/P = 2/3$
2. Therefore the final answer is $\frac{2}{3}$.

Final answer

$$\frac{2}{3}$$

Q115 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$.

Solution

1. $S/P = (-2)/(-1/3) = 6$
2. Therefore the final answer is 6.

Final answer

$$6$$

Worked Solutions

Quadratic Equations

Q116 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\frac{2}{\alpha} + \frac{2}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is -4 .

Correct option: C**Final answer** -4

Q117 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\frac{\beta}{\alpha} + \frac{\alpha}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{2}{3}$.

Correct option: D**Final answer** $\frac{2}{3}$

Worked Solutions

Quadratic Equations

Q118 MCQ

Find the sum and product of the roots of $5x^2 - 3 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $S = 0$, $P = -\frac{3}{5}$.

Correct option: D**Final answer**

$$S = 0, \quad P = -\frac{3}{5}$$

Q119 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{2}{3}$.

Correct option: D**Final answer**

$$\frac{2}{3}$$

Worked Solutions

Quadratic Equations

Q120 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 6.

Correct option: D**Final answer**

6

Q121 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $(\alpha + 1)(\beta + 1)$.

Solution

1. $P + S + 1 = -3/2 + 3 + 1 = 5/2$
2. Therefore the final answer is $\frac{5}{2}$.

Final answer $\frac{5}{2}$

Worked Solutions

Quadratic Equations

Q122 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $(\alpha - 1)(\beta - 1)$.

Solution

1. $P - S + 1 = 3/2 - 2 + 1 = 1/2$
2. Therefore the final answer is $\frac{1}{2}$.

Final answer

$$\frac{1}{2}$$

Q123 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $(\alpha + 2)(\beta + 2)$.

Solution

1. $P + 2S + 4 = 3 + 4 + 4 = 11$
2. Therefore the final answer is 11.

Final answer

$$11$$

Worked Solutions

Quadratic Equations

Q124 **Written**

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $(\alpha - 2)(\beta - 2)$.

Solution

1. $P - 2S + 4 = -1/3 + 4 + 4 = 23/3$
2. Therefore the final answer is $\frac{23}{3}$.

Final answer

$$\frac{23}{3}$$

Q125 **MCQ**

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $(\alpha + 1)(\beta + 1)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{5}{2}$.

Correct option: A**Final answer**

$$\frac{5}{2}$$

Worked Solutions

Quadratic Equations

Q126 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $(\alpha - 1)(\beta - 1)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{1}{2}$.

Correct option: C

Final answer

$$\frac{1}{2}$$

Q127 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $(\alpha + 2)(\beta + 2)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 11.

Correct option: D

Final answer

$$11$$

Worked Solutions

Quadratic Equations

Q128 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $(\alpha - 2)(\beta - 2)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{23}{3}$.

Correct option: D

Final answer

$$\frac{23}{3}$$

Q129 Written

If one root of $x^2 - 8x + k = 0$ is three times the other, find k and the roots.

Solution

1. Let roots $\alpha, 3\alpha$. $4\alpha = 8 \Rightarrow \alpha = 2$.
2. $k = \alpha(3\alpha) = 6$
3. Therefore the final answer is $k = 6$, $x = 2, 6$.

Final answer

$$k = 6, \quad x = 2, 6$$

Worked Solutions

Quadratic Equations

Q130 Written

If one root of $x^2 + 6x + m = 0$ is twice the other, find m and the roots.

Solution

1. Roots $\alpha, 2\alpha$; $3\alpha = -6 \Rightarrow \alpha = -2$; $m = 2\alpha^2 = 8$
2. Therefore the final answer is $m = 8$, $x = -2, -4$.

Final answer

$$m = 8, \quad x = -2, -4$$

Q131 Written

If one root of $8x^2 - mx + 1 = 0$ is twice the other, find m and the roots.

Solution

1. Roots $\alpha, 2\alpha$; $2\alpha^2 = 1/8 \Rightarrow \alpha = \pm 1/4$.
2. $m = 8(3\alpha) = \pm 6$.
3. Therefore the final answer is $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$.

Final answer

$$(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2}) \text{ or } (-6, -\frac{1}{4}, -\frac{1}{2})$$

Worked Solutions

Quadratic Equations

Q132 MCQ

If one root of $x^2 - 8x + k = 0$ is three times the other, find k and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $k = 6$, $x = 2, 6$.

Correct option: A**Final answer**

$$k = 6, \quad x = 2, 6$$

Q133 MCQ

If one root of $x^2 + 6x + m = 0$ is twice the other, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $m = 8$, $x = -2, -4$.

Correct option: A**Final answer**

$$m = 8, \quad x = -2, -4$$

Worked Solutions

Quadratic Equations

Q134 MCQ

If one root of $8x^2 - mx + 1 = 0$ is twice the other, find m and the roots. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$.

Correct option: D

Final answer

$$(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2}) \text{ or } (-6, -\frac{1}{4}, -\frac{1}{2})$$

Worked Solutions

Quadratic Equations

Q135 Written

If the difference between the two roots of $x^2 - (m - 1)x - 6 = 0$ is 5, find m and the roots.

Solution

1. Let roots be α and $\alpha + 5$.
2. $\alpha(\alpha + 5) = -6 \Rightarrow \alpha = -2$ or -3 .
3. $S = 2\alpha + 5 = m - 1$ gives $m = 2$ or 0 ; obtain the paired roots.
4. Therefore the final answer is $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$.

Final answer

$$(m, x_1, x_2) = (2, -2, 3) \text{ or } (0, -3, 2)$$

Q136 Written

If the difference between the roots of $x^2 - (m + 1)x + 2 = 0$ is 1, find m and the roots.

Solution

1. Let roots be α and $\alpha + 1$. $\alpha(\alpha + 1) = 2 \Rightarrow \alpha = 1$ or -2 .
2. $m + 1 = 2\alpha + 1 \Rightarrow m = 2$ or -4 .
3. Therefore the final answer is $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$.

Final answer

$$(m, x_1, x_2) = (2, 1, 2) \text{ or } (-4, -2, -1)$$

Worked Solutions

Quadratic Equations

Q137 Written

If the difference between the roots of $x^2 + 6x + m = 0$ is 4, find m and the roots.

Solution

1. Roots $\alpha, \alpha + 4$; $2\alpha + 4 = -6 \rightarrow \alpha = -5$; $m = \alpha(\alpha + 4) = 5$
2. Therefore the final answer is $m = 5$, $x = -5, -1$.

Final answer

$$m = 5, \quad x = -5, -1$$

Q138 Written

If the difference between the roots of $4x^2 + mx + 3 = 0$ is 1, find m and the roots.

Solution

1. Roots $\alpha, \alpha + 1$; $\alpha(\alpha + 1) = 3/4 \rightarrow \alpha = 1/2 \text{ or } -3/2$.
2. $-m/4 = 2\alpha + 1 \rightarrow m = -8 \text{ or } 8$.
3. Therefore the final answer is $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$.

Final answer

$$(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2}) \text{ or } (8, -\frac{3}{2}, -\frac{1}{2})$$

Worked Solutions

Quadratic Equations

Q139 MCQ

If the difference between the two roots of $x^2 - (m - 1)x - 6 = 0$ is 5, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$.

Correct option: D

Final answer

$$(m, x_1, x_2) = (2, -2, 3) \text{ or } (0, -3, 2)$$

Worked Solutions

Quadratic Equations

Q140 MCQ

If the difference between the roots of $x^2 - (m + 1)x + 2 = 0$ is 1, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$.

Correct option: A

Final answer

$(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$

Worked Solutions

Quadratic Equations

Q141 MCQ

If the difference between the roots of $x^2 + 6x + m = 0$ is 4, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $m = 5$, $x = -5, -1$.

Correct option: B

Final answer

$$m = 5, \quad x = -5, -1$$

Worked Solutions

Quadratic Equations

Q142 MCQ

If the difference between the roots of $4x^2 + mx + 3 = 0$ is 1, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$.

Correct option: B

Final answer

$$(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2}) \text{ or } (8, -\frac{3}{2}, -\frac{1}{2})$$

Worked Solutions

Quadratic Equations

Q143 Written

If the ratio of the roots of $x^2 - 2x + m = 0$ is 2 : 3, find m and the roots.

Solution

- Let roots $2t, 3t$; $5t = 2 \rightarrow t = 2/5$; $m = 6t^2 = 24/25$
- Therefore the final answer is $m = \frac{24}{25}$, $x = \frac{4}{5}, \frac{6}{5}$.

Final answer

$$m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$$

Q144 MCQ

If the ratio of the roots of $x^2 - 2x + m = 0$ is 2 : 3, find m and the roots Select the correct answer:

Solution

- Apply the same quadratic method as the given demand.
- The correct value is $m = \frac{24}{25}$, $x = \frac{4}{5}, \frac{6}{5}$.

Correct option: A

Final answer

$$m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$$

Worked Solutions

Quadratic Equations

Q145 Written

If one root of $x^2 - 6x + k = 0$ is the square of the other, find k and the roots.

Solution

1. Roots α, α^2 ; $\alpha + \alpha^2 = 6 \Rightarrow \alpha = 2$ or -3 .
2. $k = \alpha^3 = 8$ or -27 .
3. Therefore the final answer is $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$.

Final answer

$$(k, x_1, x_2) = (8, 2, 4) \text{ or } (-27, -3, 9)$$

Worked Solutions

Quadratic Equations

Q146 MCQ

If one root of $x^2 - 6x + k = 0$ is the square of the other, find k and the roots. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$.

Correct option: B

Final answer

$$(k, x_1, x_2) = (8, 2, 4) \text{ or } (-27, -3, 9)$$

Worked Solutions

Quadratic Equations

Q147 Written

Factorise $2x^2 - 3x + 2$ over the complex numbers.

Solution

1. $D = 9 - 16 = -7$; roots $(3 \pm i\sqrt{7})/4$
2. Keep leading coefficient 2
3. Therefore the final answer is $2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$.

Final answer

$$2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$$

Q148 Written

Factorise $x^2 + 2x + 5$ over the complex numbers.

Solution

1. Roots $-1 \pm 2i$
2. Therefore the final answer is $(x + 1 - 2i)(x + 1 + 2i)$.

Final answer

$$(x + 1 - 2i)(x + 1 + 2i)$$

Worked Solutions

Quadratic Equations

Q149 Written

Factorise $2x^2 - 6x + 1$ over the complex numbers.

Solution

1. $D = 28$; roots $(3 \pm \sqrt{7})/2$
2. Keep leading coefficient 2
3. Therefore the final answer is $2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right)$.

Final answer

$$2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right)$$

Q150 Written

Factorise $x^2 + 4x + 5$ over the complex numbers.

Solution

1. $D = 16 - 20 = -4$; roots $-2 \pm i$
2. Therefore the final answer is $(x + 2 - i)(x + 2 + i)$.

Final answer

$$(x + 2 - i)(x + 2 + i)$$

Worked Solutions

Quadratic Equations

Q151 **Written****Factorise $x^2 + 4x + 2$ over the complex numbers.****Solution**

1. *Roots* $-2 \pm \sqrt{2}$
2. Therefore the final answer is $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$.

Final answer

$$(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$$

Q152 **Written****Factorise $x^2 - 4x + 7$ over the complex numbers.****Solution**

1. $D = 16 - 28 = -12$; *roots* $2 \pm i\sqrt{3}$
2. Therefore the final answer is $(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$.

Final answer

$$(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$$

Worked Solutions

Quadratic Equations

Q153 Written

Factorise $2x^2 - x + 1$ over the complex numbers.**Solution**

1. $D = 1 - 8 = -7$; roots $(1 \pm i\sqrt{7})/4$
2. Therefore the final answer is $2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$.

Final answer

$$2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$$

Q154 Written

Factorise $3x^2 - 5x + 3$ over the complex numbers.**Solution**

1. $D = 25 - 36 = -11$; roots $(5 \pm i\sqrt{11})/6$
2. Therefore the final answer is $3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$.

Final answer

$$3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$$

Worked Solutions

Quadratic Equations

Q155 **Written****Factorise $2x^2 - 2x - 1$ over the complex numbers.****Solution**

1. $D = 4 + 8 = 12$; roots $(1 \pm \sqrt{3})/2$

2. Therefore the final answer is $2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right)$.

Final answer

$$2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right)$$

Worked Solutions

Quadratic Equations

Q156 MCQ

Factorise $2x^2 - 3x + 2$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$.

Correct option: C**Final answer**

$$2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$$

Worked Solutions

Quadratic Equations

Q157 MCQ

Factorise $x^2 + 2x + 5$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x + 1 - 2i)(x + 1 + 2i)$.

Correct option: A**Final answer**

$$(x + 1 - 2i)(x + 1 + 2i)$$

Worked Solutions

Quadratic Equations

Q158 MCQ

Factorise $2x^2 - 6x + 1$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right)$.

Correct option: D**Final answer**

$$2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right)$$

Worked Solutions

Quadratic Equations

Q159 MCQ

Factorise $x^2 + 4x + 5$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x + 2 - i)(x + 2 + i)$.

Correct option: D**Final answer**

$$(x + 2 - i)(x + 2 + i)$$

Q160 MCQ

Factorise $x^2 + 4x + 2$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$.

Correct option: B**Final answer**

$$(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$$

Worked Solutions

Quadratic Equations

Q161 MCQ

Factorise $x^2 - 4x + 7$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$.

Correct option: A**Final answer**

$$(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$$

Worked Solutions

Quadratic Equations

Q162 MCQ

Factorise $2x^2 - x + 1$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$.**Correct option: C****Final answer**

$$2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$$

Worked Solutions

Quadratic Equations

Q163 MCQ

Factorise $3x^2 - 5x + 3$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$.

Correct option: A**Final answer**

$$3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$$

Worked Solutions

Quadratic Equations

Q164 MCQ

Factorise $2x^2 - 2x - 1$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right)$.

Correct option: B**Final answer**

$$2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right)$$

Worked Solutions

Quadratic Equations

Q165 **Written****Factorise $x^4 + 2x^2 - 3$ over the complex numbers.****Solution**

1. $x^4 + 2x^2 - 3 = (x^2 - 1)(x^2 + 3)$
2. Factor each quadratic
3. Therefore the final answer is $(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{3})$.

Final answer

$$(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{3})$$

Q166 **Written****Factorise $x^4 - x^2 - 6$ over the complex numbers.****Solution**

1. Let $y = x^2$: $y^2 - y - 6 = (y - 3)(y + 2)$
2. Factor $x^2 - 3$ and $x^2 + 2$
3. Therefore the final answer is $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{2})$.

Final answer

$$(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{2})$$

Worked Solutions

Quadratic Equations

Q167 Written

Factorise $x^4 - 9$ over the complex numbers.

Solution

1. $x^4 - 9 = (x^2 - 3)(x^2 + 3)$
2. Factor each quadratic
3. Therefore the final answer is $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$.

Final answer

$$(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$$

Q168 Written

Factorise $x^4 - 3x^2 + 2$ over the complex numbers.

Solution

1. Let $y = x^2$: $y^2 - 3y + 2 = (y - 1)(y - 2)$
2. Factor $x^2 - 1$ and $x^2 - 2$
3. Therefore the final answer is $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$.

Final answer

$$(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$$

Worked Solutions

Quadratic Equations

Q169 MCQ

Factorise $x^4 + 2x^2 - 3$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{3})$.

Correct option: B**Final answer**

$$(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{3})$$

Q170 MCQ

Factorise $x^4 - x^2 - 6$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{2})$.

Correct option: B**Final answer**

$$(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{2})$$

Worked Solutions

Quadratic Equations

Q171 MCQ

Factorise $x^4 - 9$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$.

Correct option: D**Final answer**

$$(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$$

Q172 MCQ

Factorise $x^4 - 3x^2 + 2$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$.

Correct option: C**Final answer**

$$(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$$

Worked Solutions

Quadratic Equations

Q173 Written

Create a quadratic equation with integer coefficients having $\frac{1+i}{2}$ and $\frac{1-i}{2}$ as roots.

Solution

1. $S = 1, P = 1/2$
2. $x^2 - x + 1/2 = 0$; multiply by 2
3. Therefore the final answer is $2x^2 - 2x + 1 = 0$.

Final answer

$$2x^2 - 2x + 1 = 0$$

Q174 Written

Find two numbers whose sum is -1 and product is 2 .

Solution

1. They are roots of $x^2 + x + 2 = 0$
2. $D = -7$
3. Therefore the final answer is $\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$.

Final answer

$$\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$$

Worked Solutions

Quadratic Equations

Q175 Written

Create a quadratic equation with roots $3 + 2i$ and $3 - 2i$.

Solution

1. $S = 6, P = 13$
2. $x^2 - Sx + P = 0$
3. Therefore the final answer is $x^2 - 6x + 13 = 0$.

Final answer

$$x^2 - 6x + 13 = 0$$

Q176 Written

Find two numbers whose sum and product are both 1.

Solution

1. Roots of $x^2 - x + 1 = 0$
2. $D = -3$
3. Therefore the final answer is $\frac{1 + i\sqrt{3}}{2}, \frac{1 - i\sqrt{3}}{2}$.

Final answer

$$\frac{1 + i\sqrt{3}}{2}, \frac{1 - i\sqrt{3}}{2}$$

Worked Solutions

Quadratic Equations

Q177 Written

Create a quadratic equation with roots -1 and 2 .**Solution**

1. $S = 1, P = -2$
2. Therefore the final answer is $x^2 - x - 2 = 0$.

Final answer

$$x^2 - x - 2 = 0$$

Q178 Written

Create a quadratic equation with roots $1 + 4i$ and $1 - 4i$.**Solution**

1. $S = 2, P = 17$
2. Therefore the final answer is $x^2 - 2x + 17 = 0$.

Final answer

$$x^2 - 2x + 17 = 0$$

Worked Solutions

Quadratic Equations

Q179 Written

Create an integer-coefficient quadratic with roots $\frac{5+\sqrt{7}}{3}$ and $\frac{5-\sqrt{7}}{3}$.

Solution

1. $S = 10/3, P = 2$
2. $x^2 - (10/3)x + 2 = 0$; multiply by 3
3. Therefore the final answer is $3x^2 - 10x + 6 = 0$.

Final answer

$$3x^2 - 10x + 6 = 0$$

Q180 Written

Find two numbers whose sum is 2 and product is 5.

Solution

1. Roots of $x^2 - 2x + 5 = 0$
2. $D = -16$
3. Therefore the final answer is $1 + 2i, 1 - 2i$.

Final answer

$$1 + 2i, 1 - 2i$$

Worked Solutions

Quadratic Equations

Q181 **Written****Find two numbers whose sum is 4 and product is 2.****Solution**

1. Roots of $x^2 - 4x + 2 = 0$
2. Therefore the final answer is $2 + \sqrt{2}$, $2 - \sqrt{2}$.

Final answer

$$2 + \sqrt{2}, 2 - \sqrt{2}$$

Q182 **MCQ****Create a quadratic equation with integer coefficients having $\frac{1+i}{2}$ and $\frac{1-i}{2}$ as roots Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $2x^2 - 2x + 1 = 0$.

Correct option: B**Final answer**

$$2x^2 - 2x + 1 = 0$$

Worked Solutions

Quadratic Equations

Q183 MCQ

Find two numbers whose sum is -1 and product is 2 Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$.**Correct option: B****Final answer**

$$\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$$

Q184 MCQ

Create a quadratic equation with roots $3 + 2i$ and $3 - 2i$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x^2 - 6x + 13 = 0$.**Correct option: B****Final answer**

$$x^2 - 6x + 13 = 0$$

Worked Solutions

Quadratic Equations

Q185 MCQ

Find two numbers whose sum and product are both 1 Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $\frac{1 + i\sqrt{3}}{2}, \frac{1 - i\sqrt{3}}{2}$.**Correct option: A****Final answer**

$$\frac{1 + i\sqrt{3}}{2}, \frac{1 - i\sqrt{3}}{2}$$

Q186 MCQ

Create a quadratic equation with roots -1 and 2 Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x^2 - x - 2 = 0$.**Correct option: A****Final answer**

$$x^2 - x - 2 = 0$$

Worked Solutions

Quadratic Equations

Q187 MCQ

Create a quadratic equation with roots $1 + 4i$ and $1 - 4i$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $x^2 - 2x + 17 = 0$.

Correct option: D

Final answer

$$x^2 - 2x + 17 = 0$$

Q188 MCQ

Create an integer-coefficient quadratic with roots $\frac{5+\sqrt{7}}{3}$ and $\frac{5-\sqrt{7}}{3}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $3x^2 - 10x + 6 = 0$.

Correct option: A

Final answer

$$3x^2 - 10x + 6 = 0$$

Worked Solutions

Quadratic Equations

Q189 MCQ

Find two numbers whose sum is 2 and product is 5 Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $1 + 2i$, $1 - 2i$.

Correct option: B**Final answer** $1 + 2i, 1 - 2i$

Q190 MCQ

Find two numbers whose sum is 4 and product is 2 Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $2 + \sqrt{2}$, $2 - \sqrt{2}$.

Correct option: B**Final answer** $2 + \sqrt{2}, 2 - \sqrt{2}$

Worked Solutions

Quadratic Equations

Q191 Written

If α, β are roots of $x^2 + 2x - 4 = 0$, find a quadratic with coefficient of x^2 equal to 1 and roots $\alpha + 2, \beta + 2$.

Solution

1. $S = -2, P = -4$
2. $\text{Newsum} = S + 4 = 2; \text{newproduct} = P + 2S + 4 = -4$
3. $\text{Equation } x^2 - 2x - 4 = 0$
4. Therefore the final answer is $x^2 - 2x - 4 = 0$.

Final answer

$$x^2 - 2x - 4 = 0$$

Q192 MCQ

If α, β are roots of $x^2 + 2x - 4 = 0$, find a quadratic with coefficient of x^2 equal to 1 and roots $\alpha + 2, \beta + 2$. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $x^2 - 2x - 4 = 0$.

Correct option: C

Final answer

$$x^2 - 2x - 4 = 0$$

Worked Solutions

Quadratic Equations

Q193 Written

If α, β are roots of $x^2 + x - 3 = 0$, find a quadratic with roots α^2, β^2 .

Solution

1. $S = -1, P = -3$
2. $\text{Newsum} S^2 - 2P = 1 + 6 = 7$; $\text{product} P^2 = 9$
3. Therefore the final answer is $x^2 - 7x + 9 = 0$.

Final answer

$$x^2 - 7x + 9 = 0$$

Q194 Written

If α, β are roots of $x^2 - 2x + 7 = 0$, find a quadratic with roots α^2, β^2 .

Solution

1. $S = 2, P = 7$
2. $\text{Newsum} = S^2 - 2P = -10$; $\text{product} = P^2 = 49$
3. Therefore the final answer is $x^2 + 10x + 49 = 0$.

Final answer

$$x^2 + 10x + 49 = 0$$

Worked Solutions

Quadratic Equations

Q195 MCQ

If α, β are roots of $x^2 + x - 3 = 0$, find a quadratic with roots α^2, β^2 . Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $x^2 - 7x + 9 = 0$.

Correct option: D**Final answer**

$$x^2 - 7x + 9 = 0$$

Q196 MCQ

If α, β are roots of $x^2 - 2x + 7 = 0$, find a quadratic with roots α^2, β^2 . Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $x^2 + 10x + 49 = 0$.

Correct option: C**Final answer**

$$x^2 + 10x + 49 = 0$$

Worked Solutions

Quadratic Equations

Q197 Written

If α, β are roots of $2x^2 + 3x + 4 = 0$, find an integer-coefficient quadratic with roots $\frac{1}{\alpha}, \frac{1}{\beta}$.

Solution

1. $S = -3/2, P = 2$
2. $\text{Newsum} = S/P = -3/4$; $\text{product} = 1/P = 1/2$
3. $x^2 + (3/4)x + 1/2 = 0$; multiply by 4
4. Therefore the final answer is $4x^2 + 3x + 2 = 0$.

Final answer

$$4x^2 + 3x + 2 = 0$$

Q198 MCQ

If α, β are roots of $2x^2 + 3x + 4 = 0$, find an integer-coefficient quadratic with roots $\frac{1}{\alpha}, \frac{1}{\beta}$. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $4x^2 + 3x + 2 = 0$.

Correct option: A

Final answer

$$4x^2 + 3x + 2 = 0$$

Worked Solutions

Quadratic Equations

Q199 Written

If α, β are roots of $x^2 + 2x + 3 = 0$, find a quadratic with roots $\alpha + 3\beta$ and $3\alpha + \beta$.

Solution

1. $S = -2, P = 3$
2. $\text{Newsum} = 4S = -8$
3. $\text{Newproduct} = 3(\alpha^2 + \beta^2) + 10P = 3(S^2 - 2P) + 10P = 24$
4. Therefore the final answer is $x^2 + 8x + 24 = 0$.

Final answer

$$x^2 + 8x + 24 = 0$$

Q200 MCQ

If α, β are roots of $x^2 + 2x + 3 = 0$, find a quadratic with roots $\alpha + 3\beta$ and $3\alpha + \beta$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $x^2 + 8x + 24 = 0$.

Correct option: D

Final answer

$$x^2 + 8x + 24 = 0$$

Worked Solutions

Quadratic Equations

Q201 Written

If $x^2 + 2mx + m + 2 = 0$ has two distinct positive roots, find the range of m .

Solution

1. $D > 0 \rightarrow (m - 2)(m + 1) > 0 \rightarrow m < -1 \text{ or } m > 2$
2. $S = -2m > 0 \rightarrow m < 0$
3. $P = m + 2 > 0 \rightarrow m > -2$
4. Intersection : $-2 < m < -1$
5. Therefore the final answer is $-2 < m < -1$.

Final answer

$$-2 < m < -1$$

Worked Solutions

Quadratic Equations

Q202 Written

If $x^2 + 2(m - 1)x + 2m^2 - 5m - 3 = 0$ has two distinct positive roots, find the range of m .

Solution

1. $D = 4(4 - m)(m + 1) > 0 \rightarrow -1 < m < 4$
2. $S = 2(1 - m) > 0 \rightarrow m < 1$
3. $P = (2m + 1)(m - 3) > 0 \rightarrow m < -1/2 \text{ or } m > 3$
4. Intersection : $-1 < m < -1/2$
5. Therefore the final answer is $-1 < m < -\frac{1}{2}$.

Final answer

$$-1 < m < -\frac{1}{2}$$

Worked Solutions

Quadratic Equations

Q203 MCQ

If $x^2 + 2mx + m + 2 = 0$ has two distinct positive roots, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-2 < m < -1$.

Correct option: A**Final answer**

$$-2 < m < -1$$

Q204 MCQ

If $x^2 + 2(m - 1)x + 2m^2 - 5m - 3 = 0$ has two distinct positive roots, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-1 < m < -\frac{1}{2}$.

Correct option: B**Final answer**

$$-1 < m < -\frac{1}{2}$$

Worked Solutions

Quadratic Equations

Q205 **Written**

If $x^2 - 2mx + m + 6 = 0$ has two distinct negative roots, find the range of m .

Solution

1. $D > 0 : 4m^2 - 4(m + 6) > 0 \rightarrow (m - 3)(m + 2) > 0$
2. $S = 2m < 0 \rightarrow m < 0$
3. $P = m + 6 > 0 \rightarrow m > -6$
4. *Intersection* : $-6 < m < -2$
5. Therefore the final answer is $-6 < m < -2$.

Final answer

$$-6 < m < -2$$

Worked Solutions

Quadratic Equations

Q206 Written

If $x^2 + kx + 2k + 5 = 0$ has two distinct negative roots, find the range of k .

Solution

1. $D = (k - 10)(k + 2) > 0 \rightarrow k < -2 \text{ or } k > 10$
2. $S = -k < 0 \rightarrow k > 0$
3. $P = 2k + 5 > 0 \rightarrow k > -5/2$
4. Intersection $k > 10$
5. Therefore the final answer is $k > 10$.

Final answer

$$k > 10$$

Worked Solutions

Quadratic Equations

Q207 MCQ

If $x^2 - 2mx + m + 6 = 0$ has two distinct negative roots, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-6 < m < -2$.

Correct option: C**Final answer**

$$-6 < m < -2$$

Q208 MCQ

If $x^2 + kx + 2k + 5 = 0$ has two distinct negative roots, find the range of k Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $k > 10$.

Correct option: B**Final answer**

$$k > 10$$

Worked Solutions

Quadratic Equations

Q209 **Written**

If $x^2 - (m - 3)x - m - 4 = 0$ has roots of different signs, find the range of m .

Solution

1. *Opposite signs if* $P = -m - 4 < 0$
2. Therefore the final answer is $m > -4$.

Final answer

$$m > -4$$

Q210 **Written**

If $x^2 + 4kx - k - 3 = 0$ has roots of different signs, find the range of k .

Solution

1. $P = -k - 3 < 0$
2. Therefore the final answer is $k > -3$.

Final answer

$$k > -3$$

Worked Solutions

Quadratic Equations

Q211 MCQ

If $x^2 - (m - 3)x - m - 4 = 0$ has roots of different signs, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $m > -4$.

Correct option: A**Final answer**

$$m > -4$$

Q212 MCQ

If $x^2 + 4kx - k - 3 = 0$ has roots of different signs, find the range of k Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $k > -3$.

Correct option: D**Final answer**

$$k > -3$$

Worked Solutions

Quadratic Equations

Q213 Challenge

For $x^2 + ax - a = 0$, determine the nature of the solutions as a varies.

Solution

1. $D = a^2 + 4a = a(a + 4)$
2. Classify sign intervals at -4 and 0
3. Therefore the final answer is $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary.

Final answer

$a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary

Worked Solutions

Quadratic Equations

Q214 Challenge

For $x^2 - (k + 2)x + k^2 = 0$, determine the nature of the solutions as k varies.

Solution

1. $D = (k + 2)^2 - 4k^2 = -3k^2 + 4k + 4 = -3(k - 2)(k + 2/3)$
2. Classify D sign
3. Therefore the final answer is $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary.

Final answer

$-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary

Worked Solutions

Quadratic Equations

Q215 Challenge

For $x^2 - 2(a - 1)x + a^2 - 3a + 4 = 0$, determine the nature of the solutions as a varies.

Solution

1. $D = 4(a - 1)^2 - 4(a^2 - 3a + 4) = 4(a - 3)$
2. Classify by sign of $a - 3$
3. Therefore the final answer is $a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary.

Final answer

$a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary

Worked Solutions

Quadratic Equations

Q216 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^2\beta + \alpha\beta^2$.

Solution

1. $\alpha\beta(\alpha + \beta) = PS = (3/2)(3) = 9/2$
2. Therefore the final answer is $\frac{9}{2}$.

Final answer

$$\frac{9}{2}$$

Worked Solutions

Quadratic Equations

Q217 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $(\alpha - \beta)^2$.

Solution

1. $S^2 - 4P = 9 - 6 = 3$
2. Therefore the final answer is 3.

Final answer

3

Worked Solutions

Quadratic Equations

Q218 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 3, P = 3/2$
2. $S^3 - 3PS = 27 - 27/2 = 27/2$
3. Therefore the final answer is $\frac{27}{2}$.

Final answer

$$\frac{27}{2}$$

Worked Solutions

Quadratic Equations

Q219 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\frac{\beta^2}{\alpha} + \frac{\alpha^2}{\beta}$.

Solution

1. $(\alpha^3 + \beta^3)/(\alpha\beta) = (27/2)/(3/2) = 9$
2. Therefore the final answer is 9.

Final answer

9

Worked Solutions

Quadratic Equations

Q220 Challenge

The quadratics $x^2 + ax + bc = 0$ and $x^2 + bx + ca = 0$ have a non-zero common root, with $a \neq b$ and $c \neq 0$. Show that the other two roots are roots of $x^2 + cx + ab = 0$.

Solution

1. *Subtract equations* : $(a - b)(x - c) = 0$, so *common root* $r = c$.
2. *Substitute* c : $c(c + a + b) = 0$; $c \neq 0 \rightarrow a + b + c = 0$.
3. *The other roots are* $bc/c = b$ and $ca/c = a$. *Their equation is* $x^2 - (a + b)x + ab = x^2 + cx + ab$.
4. Therefore the final answer is Common root c ; other roots a and b .

Final answer

Common root c ; other roots a and b .

Worked Solutions

Quadratic Equations

Q221 Challenge

For $S(x) = 6x^4 - 7x^2 - 3$, determine the possible values of x and interpret the real values in the stress model.

Solution

1. Let $y = x^2$: $6y^2 - 7y - 3 = (2y - 3)(3y + 1)$
2. $x^2 = 3/2$ or $-1/3$; hence $x = \pm\sqrt{6}/2$ or $\pm i\sqrt{3}/3$
3. For real displacement x , only $\pm\sqrt{6}/2$ are admissible
4. Therefore the final answer is $x = \pm\frac{\sqrt{6}}{2}$, $x = \pm\frac{i\sqrt{3}}{3}$; real: $\pm\sqrt{6}/2$.

Final answer

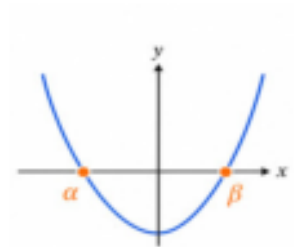
$$x = \pm\frac{\sqrt{6}}{2}, \quad x = \pm\frac{i\sqrt{3}}{3}; \text{ real: } \pm\sqrt{6}/2$$

Worked Solutions

Quadratic Equations

Q222 Challenge

In the opposite figure, α, β are the two roots. If α, β are non-zero roots of a quadratic, construct a quadratic whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$.



Solution

1. $\text{Newsum} = S/P; \text{newproduct} = 1/P$
2. Equation $x^2 - (S/P)x + 1/P = 0$; multiply by P
3. Therefore the final answer is For $P = \alpha\beta \neq 0$: $Px^2 - Sx + 1 = 0$.

Final answer

For $P = \alpha\beta \neq 0$: $Px^2 - Sx + 1 = 0$

Worked Solutions

Quadratic Equations

Q223 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both positive.

Solution

1. $D = (m + 1)^2 - 4(m + 4) > 0 \rightarrow m < -3 \text{ or } m > 5$
2. $S = -(m + 1) > 0 \rightarrow m < -1$
3. $P = m + 4 > 0 \rightarrow m > -4$
4. Intersection $(-4, -3)$
5. Therefore the final answer is $-4 < m < -3$.

Final answer

$$-4 < m < -3$$

Worked Solutions

Quadratic Equations

Q224 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both negative.

Solution

1. $D > 0 \rightarrow m < -3$ or $m > 5$
2. $S < 0 \rightarrow m > -1$
3. $P > 0 \rightarrow m > -4$
4. *Intersection* $m > 5$
5. Therefore the final answer is $m > 5$.

Final answer

$$m > 5$$

Worked Solutions

Quadratic Equations

Q225 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two roots of opposite signs.

Solution

1. *Opposite signs if $P = m + 4 < 0$*
2. Therefore the final answer is $m < -4$.

Final answer

$$m < -4$$

Worked Solutions

Quadratic Equations

Q226 Challenge

A model has $R = 3x^2 + 2\sqrt{3}x + 2$. Determine the values of x for which $R = 0$, and interpret the result.

Solution

1. Set $R = 0$. $D = 12 - 24 = -12$.
2. $x = (-2\sqrt{3} \pm i2\sqrt{3})/6 = (-\sqrt{3} \pm i\sqrt{3})/3$.
3. Because x is real, the complex pair is not admissible; no real equilibrium exists.
4. Therefore the final answer is $x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium.

Final answer

$$x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}; \text{ no real equilibrium}$$

Worked Solutions

Quadratic Equations

Q227 Challenge

For $x^2 + ax - a = 0$, determine the nature of the solutions as a varies. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary.

Correct option: B

Final answer

$a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary

Worked Solutions

Quadratic Equations

Q228 Challenge

For $x^2 - (k + 2)x + k^2 = 0$, determine the nature of the solutions as k varies. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary.

Correct option: B

Final answer

$-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary

Worked Solutions

Quadratic Equations

Q229 Challenge

For $x^2 - 2(a - 1)x + a^2 - 3a + 4 = 0$, determine the nature of the solutions as a varies. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary.

Correct option: B

Final answer

$a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary

Worked Solutions

Quadratic Equations

Q230 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^2\beta + \alpha\beta^2$. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{9}{2}$.

Correct option: B

Final answer

$$\frac{9}{2}$$

Worked Solutions

Quadratic Equations

Q231 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $(\alpha - \beta)^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 3.

Correct option: D

Final answer

3

Worked Solutions

Quadratic Equations

Q232 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{27}{2}$.

Correct option: B

Final answer

$$\frac{27}{2}$$

Worked Solutions

Quadratic Equations

Q233 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\frac{\beta^2}{\alpha} + \frac{\alpha^2}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 9.

Correct option: A

Final answer

9

Worked Solutions

Quadratic Equations

Q234 Challenge

The quadratics $x^2 + ax + bc = 0$ and $x^2 + bx + ca = 0$ have a non-zero common root, with $a \neq b$ and $c \neq 0$. Show that the other two roots are roots of $x^2 + cx + ab = 0$. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is Common root c ; other roots a and b .

Correct option: C

Final answer

Common root c ; other roots a and b .

Worked Solutions

Quadratic Equations

Q235 Challenge

For $S(x) = 6x^4 - 7x^2 - 3$, determine the possible values of x and interpret the real values in the stress model. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \pm \frac{\sqrt{6}}{2}$, $x = \pm \frac{i\sqrt{3}}{3}$; real: $\pm \sqrt{6}/2$.

Correct option: A

Final answer

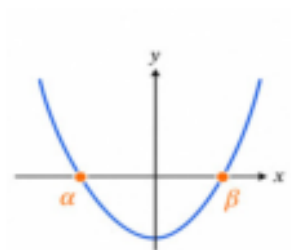
$$x = \pm \frac{\sqrt{6}}{2}, \quad x = \pm \frac{i\sqrt{3}}{3}; \text{ real: } \pm \sqrt{6}/2$$

Worked Solutions

Quadratic Equations

Q236 Challenge

If α, β are non-zero roots of a quadratic, construct a quadratic whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$. Select the correct answer:



Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is For $P = \alpha\beta \neq 0$: $Px^2 - Sx + 1 = 0$.

Correct option: A

Final answer

For $P = \alpha\beta \neq 0$: $Px^2 - Sx + 1 = 0$

Worked Solutions

Quadratic Equations

Q237 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both positive Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-4 < m < -3$.

Correct option: C

Final answer

$$-4 < m < -3$$

Worked Solutions

Quadratic Equations

Q238 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both negative. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $m > 5$.

Correct option: D

Final answer

$$m > 5$$

Worked Solutions

Quadratic Equations

Q239 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two roots of opposite signs. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $m < -4$.

Correct option: D

Final answer

$$m < -4$$

Worked Solutions

Quadratic Equations

Q240 Challenge

A model has $R = 3x^2 + 2\sqrt{3}x + 2$. Determine the values of x for which $R = 0$, and interpret the result. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium.

Correct option: D

Final answer

$$x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}; \text{ no real equilibrium}$$

Worked Solutions

Quadratic Equations

Quiz MCQ 1 Concept Quiz · MCQ

Solve $x^2 - 5x + 6 = 0$ **Select the correct answer:**

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 2, 3.

Correct option: B

Final answer

2, 3

Worked Solutions

Quadratic Equations

Quiz MCQ 2 Concept Quiz · MCQ

For $x^2 - 4x + 7 = 0$, determine the nature of the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct non-real complex roots.

Correct option: A

Final answer

Two distinct non-real complex roots

Worked Solutions

Quadratic Equations

Quiz MCQ 3 **Concept Quiz · MCQ**

If α, β are roots of $x^2 - 3x + 1 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 7.

Correct option: C

Final answer

7

Quiz MCQ 4 **Concept Quiz · MCQ**

Create a monic quadratic with roots $2 + i$ and $2 - i$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $x^2 - 4x + 5 = 0$.

Correct option: A

Final answer

$x^2 - 4x + 5 = 0$

Worked Solutions

Quadratic Equations

Quiz WRITTEN 5 Concept Quiz · Written**Solve** $2x^2 + 3x - 2 = 0$.**Solution**

1. Apply the relevant quadratic identity.
2. Final answer: $x = \frac{1}{2}, -2$

Final answer

$$x = \frac{1}{2}, -2$$

Quiz WRITTEN 6 Concept Quiz · Written**Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary roots.****Solution**

1. Apply the relevant quadratic identity.
2. Final answer: $-1 < m < 2$

Final answer

$$-1 < m < 2$$

Worked Solutions

Quadratic Equations

Quiz WRITTEN 7 Concept Quiz · Written

If α, β are roots of $x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. Apply the relevant quadratic identity.
2. Final answer: 28

Final answer

28

Quiz WRITTEN 8 Concept Quiz · Written

If α, β are roots of $x^2 - 2x + 5 = 0$, form the quadratic with roots α^2, β^2 .

Solution

1. Apply the relevant quadratic identity.
2. Final answer: $x^2 + 6x + 25 = 0$

Final answer

$x^2 + 6x + 25 = 0$

Answer key

Use the question number shown on each page.

Q001 $x = \pm 2i\sqrt{2}$

Q002 $x = \pm 2i$

Q003 $x = \pm 7i$

Q004 $x = \pm \frac{2}{3}$

Q005 $x = \pm 2\sqrt{3}$

Q006 $x = \pm \frac{i\sqrt{3}}{2}$

Q007 A

Q008 A

Q009 C

Q010 C

Q011 B

Q012 B

Q013 $x = 3 \pm 2i\sqrt{2}$

Q014 $x = 2 \pm \sqrt{5}$

Q015 $x = -6 \pm i\sqrt{6}$

Q016 $x = -4 \pm 3i\sqrt{2}$

Q017 $x = \frac{3 \pm \sqrt{7}}{2}$

Q018 A

Q019 B

Q020 C

Q021 C

Q022 D

Q023 $x = -1, \frac{9}{2}$

Q024 $x = \frac{1}{2}, -3$

Q025 $x = 0, -\frac{7}{2}$

Q026 $x = \frac{3}{2}, -4$

Q027 $x = 3, 5$

Q028 A

Q029 D

Q030 D

Q031 A

Q032 A

Q033 $x = \frac{2 \pm \sqrt{7}}{3}$

Q034 B

Q035 $x = \frac{-\sqrt{6} \pm 4i}{2}$

Q036 $x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$

Q037 $x = \frac{-1 \pm i\sqrt{3}}{2}$

Q038 $x = \frac{5 \pm i\sqrt{11}}{6}$

Q039 $x = \sqrt{3} \pm i\sqrt{2}$

Q040 $x = \frac{-\sqrt{2} \pm i}{3}$

Q041 $x = \frac{1 \pm i\sqrt{7}}{2}$

Q042 $x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$

Q043 $x = 1 \pm \frac{i\sqrt{6}}{2}$

Q044 B

Q045 C

Q046 D

Q047 B

Q048 D

Q049 B

Q050 C

Q051 C

Q052 B

Q053 Two distinct imaginary (non-real complex) solutions

Q054 $-2 \leq m \leq 6$

Q055 $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary

Q056 Two distinct imaginary (non-real complex) solutions

Q057 Two distinct real solutions

Q058 One repeated real solution

Q059 Two distinct real solutions

Q060 One repeated real solution

Q061 Two distinct imaginary (non-real complex) solutions

Q062 One repeated real solution

Q063 Two distinct imaginary (non-real complex) solutions

Q064 Two distinct real solutions

Q065 A

Q066 D

Q067 C

Q068 C

Q069 A

Q070 A

Q071 C

Q072 C

Q073 B

Q074 A

Q075 A

Q076 A

Q077 $-1 < m < 2$

Q078 $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

Q079 $a = 3$ or $a = -5$

Q080 $-18 < m < -2$

Q081 $0 \leq m \leq 3$

Q082 C

Q083 C

Q084 A

Q085 D

Q086 C

Q087 $\alpha + \beta = 2, \alpha\beta = -\frac{5}{3}$

Q088 $\alpha + \beta = -\frac{1}{2}, \alpha\beta = -\frac{7}{4}$

Q089 $S = -1, P = 2$

Q090 $S = \frac{7}{3}, P = -2$

Q091 C

Q092 A

Q093 B

Q094 A

Q095 12

Q096 $\frac{81}{2}$
 Q097 1
 Q098 -1
 Q099 -2
 Q100 -10
 Q101 $\frac{14}{3}$
 Q102 -10
 Q103 A
 Q104 B
 Q105 A
 Q106 B
 Q107 D
 Q108 B
 Q109 B
 Q110 D
 Q111 -4
 Q112 $\frac{2}{3}$
 Q113 $S = 0, \quad P = -\frac{3}{5}$
 Q114 $\frac{2}{3}$
 Q115 6
 Q116 C
 Q117 D
 Q118 D
 Q119 D
 Q120 D
 Q121 $\frac{5}{2}$
 Q122 $\frac{1}{2}$
 Q123 11
 Q124 $\frac{23}{3}$
 Q125 A

Q126 C
 Q127 D
 Q128 D
 Q129 $k = 6, \quad x = 2, 6$
 Q130 $m = 8, \quad x = -2, -4$
 Q131 $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$
 Q132 A
 Q133 A
 Q134 D
 Q135 $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$
 Q136 $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$
 Q137 $m = 5, \quad x = -5, -1$
 Q138 $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$
 Q139 D
 Q140 A
 Q141 B
 Q142 B
 Q143 $m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$
 Q144 A
 Q145 $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$
 Q146 B
 Q147 $2\left(x - \frac{3+i\sqrt{7}}{4}\right)\left(x - \frac{3-i\sqrt{7}}{4}\right)$
 Q148 $(x+1-2i)(x+1+2i)$
 Q149 $2\left(x - \frac{3+\sqrt{7}}{2}\right)\left(x - \frac{3-\sqrt{7}}{2}\right)$
 Q150 $(x+2-i)(x+2+i)$
 Q151 $(x+2-\sqrt{2})(x+2+\sqrt{2})$
 Q152 $(x-2-i\sqrt{3})(x-2+i\sqrt{3})$
 Q153 $2\left(x - \frac{1+i\sqrt{7}}{4}\right)\left(x - \frac{1-i\sqrt{7}}{4}\right)$
 Q154 $3\left(x - \frac{5+i\sqrt{11}}{6}\right)\left(x - \frac{5-i\sqrt{11}}{6}\right)$

Q155 $2\left(x - \frac{1+\sqrt{3}}{2}\right)\left(x - \frac{1-\sqrt{3}}{2}\right)$
 Q156 C
 Q157 A
 Q158 D
 Q159 D
 Q160 B
 Q161 A
 Q162 C
 Q163 A
 Q164 B
 Q165 $(x-1)(x+1)(x-i\sqrt{3})(x+i\sqrt{3})$
 Q166 $(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{2})(x+i\sqrt{2})$
 Q167 $(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{3})(x+i\sqrt{3})$
 Q168 $(x-1)(x+1)(x-\sqrt{2})(x+\sqrt{2})$
 Q169 B
 Q170 B
 Q171 D
 Q172 C
 Q173 $2x^2 - 2x + 1 = 0$
 Q174 $\frac{-1+i\sqrt{7}}{2}, \frac{-1-i\sqrt{7}}{2}$
 Q175 $x^2 - 6x + 13 = 0$
 Q176 $\frac{1+i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$
 Q177 $x^2 - x - 2 = 0$
 Q178 $x^2 - 2x + 17 = 0$
 Q179 $3x^2 - 10x + 6 = 0$
 Q180 $1+2i, 1-2i$
 Q181 $2+\sqrt{2}, 2-\sqrt{2}$
 Q182 B
 Q183 B
 Q184 B

Q185 A
 Q186 A
 Q187 D
 Q188 A
 Q189 B
 Q190 B
 Q191 $x^2 - 2x - 4 = 0$
 Q192 C
 Q193 $x^2 - 7x + 9 = 0$
 Q194 $x^2 + 10x + 49 = 0$
 Q195 D
 Q196 C
 Q197 $4x^2 + 3x + 2 = 0$
 Q198 A
 Q199 $x^2 + 8x + 24 = 0$
 Q200 D
 Q201 $-2 < m < -1$
 Q202 $-1 < m < -\frac{1}{2}$
 Q203 A
 Q204 B
 Q205 $-6 < m < -2$
 Q206 $k > 10$
 Q207 C
 Q208 B
 Q209 $m > -4$
 Q210 $k > -3$
 Q211 A
 Q212 D
 Q213 $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary
 Q214 $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary
 Q215 $a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary
 Q216 $\frac{9}{2}$

Q217 3

Q218 $\frac{27}{2}$

Q219 9

Q220 Common root c ; other roots a and b .Q221 $x = \pm \frac{\sqrt{6}}{2}$, $x = \pm \frac{i\sqrt{3}}{3}$; real: $\pm \sqrt{6}/2$ Q222 For $P = \alpha\beta \neq 0$: $Px^2 - Sx + 1 = 0$ Q223 $-4 < m < -3$ Q224 $m > 5$ Q225 $m < -4$ Q226 $x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium

Q227 B

Q228 B

Q229 B

Q230 B

Q231 D

Q232 B

Q233 A

Q234 C

Q235 A

Q236 A

Q237 C

Q238 D

Q239 D

Q240 D

Quiz MCQ 1 B

Quiz MCQ 2 A

Quiz MCQ 3 C

Quiz MCQ 4 A

Quiz WRITTEN 5 $x = \frac{1}{2}, -2$ Quiz WRITTEN 6 $-1 < m < 2$

Quiz WRITTEN 7 28

Quiz WRITTEN 8 $x^2 + 6x + 25 = 0$

Concept 3 | Polynomial Theorems

English Student Edition • Focused formulas and guided practice

Formula opener

Remainder

$$P(a) = R \text{ when divisor} = x - a$$

Factor

$$P(a) = 0 \Rightarrow (x - a) \mid P(x)$$

Roots

$$P(x) = a \prod (x - r_i)$$

Worked Solutions

Polynomial Theorems

Q001 **Written****Find the remainder when $x^3 + 2x^2 - 3x - 4$ is divided by $x + 2$.****Solution**

1. $U_{\text{sex}} = -2$
2. $P(-2) = 2$
3. Therefore the final answer is 2.

Final answer

2

Q002 **Written****Find the remainder when $3x^3 + 2x^2 - 5x + 6$ is divided by $x - 1$.****Solution**

1. $U_{\text{sex}} = 1$
2. $P(1) = 6$
3. Therefore the final answer is 6.

Final answer

6

Worked Solutions

Polynomial Theorems

Q003 MCQ

Find the remainder when $x^3 + 2x^2 - 3x - 4$ is divided by $x + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is 2.

Correct option: C**Final answer**

2

Q004 MCQ

Find the remainder when $3x^3 + 2x^2 - 5x + 6$ is divided by $x - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is 6.

Correct option: C**Final answer**

6

Worked Solutions

Polynomial Theorems

Q005 Written

Find the remainder when $3x^4 - 8x^3 - 5x^2 + 12$ is divided by $3x - 2$.**Solution**

1. Use $x = 2/3$
2. Evaluate $P(2/3) = 8$
3. Therefore the final answer is 8.

Final answer

8

Q006 Written

Find the remainder when $x^3 - 5x^2 + 4x - 1$ is divided by $2x - 1$.**Solution**

1. Use $x = 1/2$
2. $P(1/2) = -1/8$
3. Therefore the final answer is $-\frac{1}{8}$.

Final answer $-\frac{1}{8}$

Worked Solutions

Polynomial Theorems

Q007 **Written****Find the remainder when $6x^3 - x^2 + x - 2$ is divided by $3x - 2$.****Solution**

1. Use $x = 2/3$
2. $P(2/3) = 0$
3. Therefore the final answer is 0.

Final answer

0

Q008 **Written****Find the remainder when $4x^3 - 2x^2 - 4x + 3$ is divided by $2x + 1$.****Solution**

1. Use $x = -1/2$
2. $P(-1/2) = 4$
3. Therefore the final answer is 4.

Final answer

4

Worked Solutions

Polynomial Theorems

Q009 MCQ

Find the remainder when $3x^4 - 8x^3 - 5x^2 + 12$ is divided by $3x - 2$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is 8.

Correct option: D**Final answer**

8

Q010 MCQ

Find the remainder when $x^3 - 5x^2 + 4x - 1$ is divided by $2x - 1$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $-\frac{1}{8}$.

Correct option: A**Final answer** $-\frac{1}{8}$

Worked Solutions

Polynomial Theorems

Q011 MCQ

Find the remainder when $6x^3 - x^2 + x - 2$ is divided by $3x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is 0.

Correct option: D

Final answer

0

Q012 MCQ

Find the remainder when $4x^3 - 2x^2 - 4x + 3$ is divided by $2x + 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is 4.

Correct option: A

Final answer

4

Worked Solutions

Polynomial Theorems

Q013 Written

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x + 3$ is 4, find a .

Solution

1. $P(-3) = 4$
2. $9a - 35 = 4$, hence $a = 13/3$
3. Therefore the final answer is $a = 13/3$.

Final answer

$$a = 13/3$$

Q014 Written

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x - 3$ is 1, find a .

Solution

1. $P(3) = 1$
2. $9a + 37 = 1$, hence $a = -4$
3. Therefore the final answer is $a = -4$.

Final answer

$$a = -4$$

Worked Solutions

Polynomial Theorems

Q015 Written

If the remainder when $x^3 - x^2 + ax - 4$ is divided by $x + 1$ is -2 , find a .

Solution

1. $P(-1) = -2$
2. $a = -4$
3. Therefore the final answer is $a = -4$.

Final answer

$$a = -4$$

Q016 Written

If the remainder when $x^3 + 4x^2 - 5x + a$ is divided by $x + 3$ is 8 , find a .

Solution

1. $P(-3) = 8$
2. $a = -16$
3. Therefore the final answer is $a = -16$.

Final answer

$$a = -16$$

Worked Solutions

Polynomial Theorems

Q017 MCQ

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x + 3$ is 4, find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $a = 13/3$.

Correct option: B**Final answer**

$$a = 13/3$$

Q018 MCQ

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x - 3$ is 1, find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $a = -4$.

Correct option: B**Final answer**

$$a = -4$$

Worked Solutions

Polynomial Theorems

Q019 MCQ

If the remainder when $x^3 - x^2 + ax - 4$ is divided by $x + 1$ is -2 , find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $a = -4$.

Correct option: C**Final answer**

$$a = -4$$

Q020 MCQ

If the remainder when $x^3 + 4x^2 - 5x + a$ is divided by $x + 3$ is 8 , find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $a = -16$.

Correct option: A**Final answer**

$$a = -16$$

Worked Solutions

Polynomial Theorems

Q021 Written

If $x^3 - 3x^2 + 4x + a$ is divisible by $x - 2$, find a .

Solution

1. $P(2) = 0$
2. $a = -4$
3. Therefore the final answer is $a = -4$.

Final answer

$$a = -4$$

Q022 MCQ

If $x^3 - 3x^2 + 4x + a$ is divisible by $x - 2$, find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $a = -4$.

Correct option: A

Final answer

$$a = -4$$

Worked Solutions

Polynomial Theorems

Q023 Written

A polynomial $P(x)$ leaves remainders 3 on division by $x - 1$ and -5 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$.

Solution

1. Let $R = ax + b$
2. $a + b = 3$ and $-3a + b = -5$
3. $R = 2x + 1$
4. Therefore the final answer is $2x + 1$.

Final answer

$$2x + 1$$

Q024 Written

A polynomial $P(x)$ leaves remainders 3 on division by $x - 2$ and -7 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$.

Solution

1. Let $R = ax + b$
2. $2a + b = 3$ and $-3a + b = -7$
3. $R = -10x + 13$
4. Therefore the final answer is $-10x + 13$.

Final answer

$$-10x + 13$$

Worked Solutions

Polynomial Theorems

Q025 **Written**

A polynomial $P(x)$ leaves remainders 5 on division by $x - 1$ and -1 on division by $x + 2$. Find the remainder on division by $(x - 1)(x + 2)$.

Solution

1. Let $R = ax + b$
2. $a + b = 5$ and $-2a + b = -1$
3. $R = 2x + 3$
4. Therefore the final answer is $2x + 3$.

Final answer

$$2x + 3$$

Q026 **Written**

A polynomial $P(x)$ leaves remainders 4 on division by $x - 2$ and -11 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$.

Solution

1. Let $R = ax + b$
2. $2a + b = 4$ and $-3a + b = -11$
3. $R = -15x + 19$
4. Therefore the final answer is $-15x + 19$.

Final answer

$$-15x + 19$$

Worked Solutions

Polynomial Theorems

Q027 MCQ

A polynomial $P(x)$ leaves remainders 3 on division by $x - 1$ and -5 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $2x + 1$.

Correct option: A

Final answer

$$2x + 1$$

Q028 MCQ

A polynomial $P(x)$ leaves remainders 3 on division by $x - 2$ and -7 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $-10x + 13$.

Correct option: A

Final answer

$$-10x + 13$$

Worked Solutions

Polynomial Theorems

Q029 MCQ

A polynomial $P(x)$ leaves remainders 5 on division by $x - 1$ and -1 on division by $x + 2$. Find the remainder on division by $(x - 1)(x + 2)$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $2x + 3$.

Correct option: A

Final answer

$$2x + 3$$

Q030 MCQ

A polynomial $P(x)$ leaves remainders 4 on division by $x - 2$ and -11 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $-15x + 19$.

Correct option: A

Final answer

$$-15x + 19$$

Worked Solutions

Polynomial Theorems

Q031 Written

A polynomial $P(x)$ leaves remainder $7x + 6$ on division by $x^2 - x - 6$, and $4x - 3$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 + x - 2$.

Solution

1. Let $R = ax + b$
2. Use shared roots -2 and 1 : $-2a + b = -8$ and $a + b = 1$
3. $R = 3x - 2$
4. Therefore the final answer is $3x - 2$.

Final answer

$$3x - 2$$

Q032 Written

A polynomial $P(x)$ leaves remainder $3x + 2$ on division by $x^2 - x - 2$, and $7x - 6$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 - 1$.

Solution

1. Let $R = ax + b$
2. At -1 and 1 : $-a + b = -1$ and $a + b = 1$
3. $R = x$
4. Therefore the final answer is x .

Final answer

$$x$$

Worked Solutions

Polynomial Theorems

Q033 Written

A polynomial $P(x)$ leaves remainder $2x - 2$ on division by $x^2 - 2x - 3$, and $-x + 7$ on division by $x^2 - 4$. Find the remainder on division by $x^2 - x - 2$.

Solution

1. Use shared roots -1 and 2
2. $-a + b = -4$ and $2a + b = 5$, so $R = 3x - 1$
3. Therefore the final answer is $3x - 1$.

Final answer

$$3x - 1$$

Q034 Written

A polynomial $P(x)$ leaves remainder $-x + 4$ on division by $x^2 - 3x + 2$, and $3x$ on division by $x^2 - 4x + 3$. Find the remainder on division by $x^2 - 5x + 6$.

Solution

1. Use shared roots 2 and 3
2. $2a + b = 2$ and $3a + b = 9$, so $R = 7x - 12$
3. Therefore the final answer is $7x - 12$.

Final answer

$$7x - 12$$

Worked Solutions

Polynomial Theorems

Q035 MCQ

A polynomial $P(x)$ leaves remainder $7x + 6$ on division by $x^2 - x - 6$, and $4x - 3$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 + x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $3x - 2$.

Correct option: B

Final answer $3x - 2$

Q036 MCQ

A polynomial $P(x)$ leaves remainder $3x + 2$ on division by $x^2 - x - 2$, and $7x - 6$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is x .

Correct option: B

Final answer x

Worked Solutions

Polynomial Theorems

Q037 MCQ

A polynomial $P(x)$ leaves remainder $2x - 2$ on division by $x^2 - 2x - 3$, and $-x + 7$ on division by $x^2 - 4$. Find the remainder on division by $x^2 - x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $3x - 1$.

Correct option: A

Final answer $3x - 1$

Q038 MCQ

A polynomial $P(x)$ leaves remainder $-x + 4$ on division by $x^2 - 3x + 2$, and $3x$ on division by $x^2 - 4x + 3$. Find the remainder on division by $x^2 - 5x + 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $7x - 12$.

Correct option: D

Final answer $7x - 12$

Worked Solutions

Polynomial Theorems

Q039 Written

A polynomial $P(x)$ leaves remainder $x - 5$ on division by $(x + 1)^2$, and 6 on division by $x - 2$. Find the remainder on division by $(x + 1)^2(x - 2)$.

Solution

1. Let $R = ax^2 + bx + c$
2. $R(-1) = -6, R'(-1) = 1, R(2) = 6$
3. Solve to get $R = x^2 + 3x - 4$
4. Therefore the final answer is $x^2 + 3x - 4$.

Final answer

$$x^2 + 3x - 4$$

Q040 MCQ

A polynomial $P(x)$ leaves remainder $x - 5$ on division by $(x + 1)^2$, and 6 on division by $x - 2$. Find the remainder on division by $(x + 1)^2(x - 2)$. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $x^2 + 3x - 4$.

Correct option: A

Final answer

$$x^2 + 3x - 4$$

Worked Solutions

Polynomial Theorems

Q041 Written

Using synthetic division, find the quotient and remainder when $x^3 + 5x + 11$ is divided by $x + 2$.

Solution

1. Use -2 and coefficients 1,0,5,11
2. Bottom row 1,-2,9,-7
3. Therefore the final answer is $Q = x^2 - 2x + 9, R = -7$.

Final answer

$$Q = x^2 - 2x + 9, R = -7$$

Q042 MCQ

Using synthetic division, find the quotient and remainder when $x^3 + 5x + 11$ is divided by $x + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $Q = x^2 - 2x + 9, R = -7$.

Correct option: A

Final answer

$$Q = x^2 - 2x + 9, R = -7$$

Worked Solutions

Polynomial Theorems

Q043 Written

Using synthetic division, find the quotient and remainder when $A = x^3 + 5x - 6$ is divided by $B = x - 1$.

Solution

1. Use 1 and coefficients 1,0,5,-6
2. Bottom row 1,1,6,0
3. Therefore the final answer is $Q = x^2 + x + 6, R = 0$.

Final answer

$$Q = x^2 + x + 6, R = 0$$

Q044 Written

Using synthetic division, find the quotient and remainder when $A = 3x^3 - 5x^2 - 3x - 1$ is divided by $B = x - 3$.

Solution

1. Use 3 and coefficients 3,-5,-3,-1
2. Bottom row 3,4,9,26
3. Therefore the final answer is $Q = 3x^2 + 4x + 9, R = 26$.

Final answer

$$Q = 3x^2 + 4x + 9, R = 26$$

Worked Solutions

Polynomial Theorems

Q045 Written

Using synthetic division, find the quotient and remainder when $A = x^3 - 5x + 6$ is divided by $B = x - 2$.

Solution

1. Use 2 and coefficients 1,0,-5,6
2. Bottom row 1,2,-1,4
3. Therefore the final answer is $Q = x^2 + 2x - 1, R = 4$.

Final answer

$$Q = x^2 + 2x - 1, R = 4$$

Q046 Written

Using synthetic division, find the quotient and remainder when $A = x^3 + 2x^2 - x - 3$ is divided by $B = x - 1$.

Solution

1. Use 1 and coefficients 1,2,-1,-3
2. Bottom row 1,3,2,-1
3. Therefore the final answer is $Q = x^2 + 3x + 2, R = -1$.

Final answer

$$Q = x^2 + 3x + 2, R = -1$$

Worked Solutions

Polynomial Theorems

Q047 **Written**

Using synthetic division, find the quotient and remainder when $A = 4x^2 - 2x + 2$ is divided by $B = x - 1$.

Solution

1. Use 1 and coefficients 4,-2,2
2. Bottom row 4,2,4
3. Therefore the final answer is $Q = 4x + 2, R = 4$.

Final answer

$$Q = 4x + 2, R = 4$$

Q048 **MCQ**

Using synthetic division, find the quotient and remainder when $A = x^3 + 5x - 6$ is divided by $B = x - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $Q = x^2 + x + 6, R = 0$.

Correct option: B

Final answer

$$Q = x^2 + x + 6, R = 0$$

Worked Solutions

Polynomial Theorems

Q049 MCQ

Using synthetic division, find the quotient and remainder when $A = 3x^3 - 5x^2 - 3x - 1$ is divided by $B = x - 3$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $Q = 3x^2 + 4x + 9, R = 26$.

Correct option: D**Final answer**

$$Q = 3x^2 + 4x + 9, R = 26$$

Q050 MCQ

Using synthetic division, find the quotient and remainder when $A = x^3 - 5x + 6$ is divided by $B = x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $Q = x^2 + 2x - 1, R = 4$.

Correct option: D**Final answer**

$$Q = x^2 + 2x - 1, R = 4$$

Worked Solutions

Polynomial Theorems

Q051 MCQ

Using synthetic division, find the quotient and remainder when $A = x^3 + 2x^2 - x - 3$ is divided by $B = x - 1$. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $Q = x^2 + 3x + 2, R = -1$.

Correct option: A

Final answer

$$Q = x^2 + 3x + 2, R = -1$$

Q052 MCQ

Using synthetic division, find the quotient and remainder when $A = 4x^2 - 2x + 2$ is divided by $B = x - 1$. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $Q = 4x + 2, R = 4$.

Correct option: B

Final answer

$$Q = 4x + 2, R = 4$$

Worked Solutions

Polynomial Theorems

Q053 Written

Using synthetic division, find the quotient and remainder when $A = 3x^4 + 5x^3 - 2x - 12$ is divided by $B = x + 2$.

Solution

1. Use -2 and coefficients 3,5,0,-2,-12
2. Bottom row 3,-1,2,-6,0
3. Therefore the final answer is $Q = 3x^3 - x^2 + 2x - 6, R = 0$.

Final answer

$$Q = 3x^3 - x^2 + 2x - 6, R = 0$$

Q054 MCQ

Using synthetic division, find the quotient and remainder when $A = 3x^4 + 5x^3 - 2x - 12$ is divided by $B = x + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $Q = 3x^3 - x^2 + 2x - 6, R = 0$.

Correct option: D

Final answer

$$Q = 3x^3 - x^2 + 2x - 6, R = 0$$

Worked Solutions

Polynomial Theorems

Q055 Written

Factorise $x^3 + 2x^2 - x - 2$.**Solution**

1. $P(1) = 0$; *divide by* $x - 1$
2. *Quotient* $x^2 + 3x + 2 = (x + 1)(x + 2)$
3. Therefore the final answer is $(x - 1)(x + 1)(x + 2)$.

Final answer

$$(x - 1)(x + 1)(x + 2)$$

Q056 Written

Factorise $x^3 - 2x^2 - 5x + 6$.**Solution**

1. $P(1) = 0$; *quotient* $x^2 - x - 6 = (x - 3)(x + 2)$
2. Completed factorisation
3. Therefore the final answer is $(x - 1)(x - 3)(x + 2)$.

Final answer

$$(x - 1)(x - 3)(x + 2)$$

Worked Solutions

Polynomial Theorems

Q057 Written

Factorise $x^3 - x^2 - 16x - 20$.**Solution**

1. $P(-2) = 0$; *divide by* $x + 2$ *twice*
2. Remaining $x - 5$
3. Therefore the final answer is $(x + 2)^2(x - 5)$.

Final answer

$$(x + 2)^2(x - 5)$$

Q058 Written

Factorise $2x^3 - x^2 - 5x - 2$.**Solution**

1. $P(2) = 0$; *quotient* $2x^2 + 3x + 1$
2. Therefore the final answer is $(x - 2)(2x + 1)(x + 1)$.

Final answer

$$(x - 2)(2x + 1)(x + 1)$$

Worked Solutions

Polynomial Theorems

Q059 Written

Factorise $x^3 - 7x - 6$.**Solution**

1. Test divisor 3 and divide
2. Remaining linear factors
3. Therefore the final answer is $(x - 3)(x + 1)(x + 2)$.

Final answer

$$(x - 3)(x + 1)(x + 2)$$

Q060 Written

Factorise $x^3 + 4x^2 + x - 6$.**Solution**

1. Test divisor 1 and divide
2. Remaining linear factors
3. Therefore the final answer is $(x - 1)(x + 2)(x + 3)$.

Final answer

$$(x - 1)(x + 2)(x + 3)$$

Worked Solutions

Polynomial Theorems

Q061 Written

Factorise $x^3 + 6x^2 + 11x + 6$.**Solution**

1. Test divisor -1 and divide
2. Remaining linear factors
3. Therefore the final answer is $(x + 1)(x + 2)(x + 3)$.

Final answer

$$(x + 1)(x + 2)(x + 3)$$

Q062 Written

Factorise $x^3 - 3x + 2$.**Solution**

1. Test divisor 1 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x - 1)^2(x + 2)$.

Final answer

$$(x - 1)^2(x + 2)$$

Worked Solutions

Polynomial Theorems

Q063 Written

Factorise $x^3 + x^2 - 8x - 12$.**Solution**

1. Test divisor -2 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x - 3)(x + 2)^2$.

Final answer

$$(x - 3)(x + 2)^2$$

Q064 Written

Factorise $x^3 - 3x^2 - 9x - 5$.**Solution**

1. Test divisor -1 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x - 5)(x + 1)^2$.

Final answer

$$(x - 5)(x + 1)^2$$

Worked Solutions

Polynomial Theorems

Q065 Written

Factorise $2x^3 - 3x^2 - 3x + 2$.**Solution**

1. Test divisor 2 and divide
2. *Factorquotient* $2x^2 + x - 1$
3. Therefore the final answer is $(x - 2)(x + 1)(2x - 1)$.

Final answer

$$(x - 2)(x + 1)(2x - 1)$$

Q066 Written

Factorise $3x^3 + x^2 - 8x + 4$.**Solution**

1. Test divisor 1 and divide
2. *Factorquotient* $3x^2 + 4x - 4$
3. Therefore the final answer is $(x - 1)(x + 2)(3x - 2)$.

Final answer

$$(x - 1)(x + 2)(3x - 2)$$

Worked Solutions

Polynomial Theorems

Q067 **Written****Factorise** $2x^3 + 7x^2 + 8x + 3$.**Solution**

1. Test divisor -1 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x + 1)^2(2x + 3)$.

Final answer

$$(x + 1)^2(2x + 3)$$

Q068 **MCQ****Factorise** $x^3 + 2x^2 - x - 2$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)(x + 1)(x + 2)$.

Correct option: D**Final answer**

$$(x - 1)(x + 1)(x + 2)$$

Worked Solutions

Polynomial Theorems

Q069 MCQ

Factorise $x^3 - 2x^2 - 5x + 6$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)(x - 3)(x + 2)$.

Correct option: D**Final answer**

$$(x - 1)(x - 3)(x + 2)$$

Q070 MCQ

Factorise $x^3 - x^2 - 16x - 20$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x + 2)^2(x - 5)$.

Correct option: C**Final answer**

$$(x + 2)^2(x - 5)$$

Worked Solutions

Polynomial Theorems

Q071 MCQ

Factorise $2x^3 - x^2 - 5x - 2$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 2)(2x + 1)(x + 1)$.

Correct option: B**Final answer**

$$(x - 2)(2x + 1)(x + 1)$$

Q072 MCQ

Factorise $x^3 - 7x - 6$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 3)(x + 1)(x + 2)$.

Correct option: B**Final answer**

$$(x - 3)(x + 1)(x + 2)$$

Worked Solutions

Polynomial Theorems

Q073 MCQ

Factorise $x^3 + 4x^2 + x - 6$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)(x + 2)(x + 3)$.

Correct option: C**Final answer**

$$(x - 1)(x + 2)(x + 3)$$

Q074 MCQ

Factorise $x^3 + 6x^2 + 11x + 6$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x + 1)(x + 2)(x + 3)$.

Correct option: B**Final answer**

$$(x + 1)(x + 2)(x + 3)$$

Worked Solutions

Polynomial Theorems

Q075 MCQ

Factorise $x^3 - 3x + 2$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)^2(x + 2)$.

Correct option: D**Final answer**

$$(x - 1)^2(x + 2)$$

Q076 MCQ

Factorise $x^3 + x^2 - 8x - 12$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 3)(x + 2)^2$.

Correct option: D**Final answer**

$$(x - 3)(x + 2)^2$$

Worked Solutions

Polynomial Theorems

Q077 MCQ

Factorise $x^3 - 3x^2 - 9x - 5$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 5)(x + 1)^2$.

Correct option: D**Final answer**

$$(x - 5)(x + 1)^2$$

Q078 MCQ

Factorise $2x^3 - 3x^2 - 3x + 2$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 2)(x + 1)(2x - 1)$.

Correct option: A**Final answer**

$$(x - 2)(x + 1)(2x - 1)$$

Worked Solutions

Polynomial Theorems

Q079 MCQ

Factorise $3x^3 + x^2 - 8x + 4$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)(x + 2)(3x - 2)$.

Correct option: C**Final answer**

$$(x - 1)(x + 2)(3x - 2)$$

Q080 MCQ

Factorise $2x^3 + 7x^2 + 8x + 3$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x + 1)^2(2x + 3)$.

Correct option: D**Final answer**

$$(x + 1)^2(2x + 3)$$

Worked Solutions

Polynomial Theorems

Q081 Written

Factorise $x^4 - 6x^3 + 8x^2 + 8x - 16$.**Solution**

1. $P(2) = 0$; divide by $x - 2$ twice
2. Remaining quadratic $x^2 - 2x - 4$
3. Therefore the final answer is $(x - 2)^2(x^2 - 2x - 4)$.

Final answer

$$(x - 2)^2(x^2 - 2x - 4)$$

Q082 Written

Factorise $x^4 + 3x^3 - 27x^2 + 13x + 42$.**Solution**

1. Test roots and divide successively
2. Four linear factors
3. Therefore the final answer is $(x - 3)(x - 2)(x + 1)(x + 7)$.

Final answer

$$(x - 3)(x - 2)(x + 1)(x + 7)$$

Worked Solutions

Polynomial Theorems

Q083 Written

Factorise $x^4 - 4x^3 + 8x^2 - 8x + 3$.**Solution**

1. $P(1) = 0$; *dividetwice*
2. *Remainingquadratic* $x^2 - 2x + 3$
3. Therefore the final answer is $(x - 1)^2(x^2 - 2x + 3)$.

Final answer

$$(x - 1)^2(x^2 - 2x + 3)$$

Q084 Written

Factorise $x^4 - 10x^3 + 35x^2 - 50x + 24$.**Solution**

1. Test roots 1,2,3,4 successively
2. Four linear factors
3. Therefore the final answer is $(x - 4)(x - 3)(x - 2)(x - 1)$.

Final answer

$$(x - 4)(x - 3)(x - 2)(x - 1)$$

Worked Solutions

Polynomial Theorems

Q085 Written

Factorise $x^4 - 5x^3 - 3x^2 + 17x - 10$.**Solution**

1. Test roots 1 and -2 successively
2. Completed factorisation
3. Therefore the final answer is $(x - 5)(x - 1)^2(x + 2)$.

Final answer

$$(x - 5)(x - 1)^2(x + 2)$$

Q086 Written

Factorise $2x^4 - 5x^3 - 20x^2 + 20x + 48$.**Solution**

1. Test roots 2, -2, 4, -3/2
2. Completed factorisation
3. Therefore the final answer is $(x - 4)(x - 2)(x + 2)(2x + 3)$.

Final answer

$$(x - 4)(x - 2)(x + 2)(2x + 3)$$

Worked Solutions

Polynomial Theorems

Q087 **Written****Factorise** $x^4 - 4x^3 - 4x^2 + 13x - 6$.**Solution**

1. Test roots 1,-2
2. Quadratic quotient remains
3. Therefore the final answer is $(x - 1)(x + 2)(x^2 - 5x + 3)$.

Final answer

$$(x - 1)(x + 2)(x^2 - 5x + 3)$$

Q088 **Written****Factorise** $x^4 - 4x^3 + 3x^2 + 2x - 6$.**Solution**

1. Test roots 3,-1
2. Quadratic quotient remains
3. Therefore the final answer is $(x - 3)(x + 1)(x^2 - 2x + 2)$.

Final answer

$$(x - 3)(x + 1)(x^2 - 2x + 2)$$

Worked Solutions

Polynomial Theorems

Q089 Written

Factorise $x^4 - x^3 + x^2 + x - 2$.**Solution**

1. Test roots 1,-1
2. Quadratic quotient remains
3. Therefore the final answer is $(x - 1)(x + 1)(x^2 - x + 2)$.

Final answer

$$(x - 1)(x + 1)(x^2 - x + 2)$$

Q090 MCQ

Factorise $x^4 - 6x^3 + 8x^2 + 8x - 16$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 2)^2(x^2 - 2x - 4)$.

Correct option: D**Final answer**

$$(x - 2)^2(x^2 - 2x - 4)$$

Worked Solutions

Polynomial Theorems

Q091 MCQ

Factorise $x^4 + 3x^3 - 27x^2 + 13x + 42$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 3)(x - 2)(x + 1)(x + 7)$.

Correct option: A**Final answer**

$$(x - 3)(x - 2)(x + 1)(x + 7)$$

Q092 MCQ

Factorise $x^4 - 4x^3 + 8x^2 - 8x + 3$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)^2(x^2 - 2x + 3)$.

Correct option: D**Final answer**

$$(x - 1)^2(x^2 - 2x + 3)$$

Worked Solutions

Polynomial Theorems

Q093 MCQ

Factorise $x^4 - 10x^3 + 35x^2 - 50x + 24$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 4)(x - 3)(x - 2)(x - 1)$.

Correct option: D**Final answer**

$$(x - 4)(x - 3)(x - 2)(x - 1)$$

Q094 MCQ

Factorise $x^4 - 5x^3 - 3x^2 + 17x - 10$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 5)(x - 1)^2(x + 2)$.

Correct option: C**Final answer**

$$(x - 5)(x - 1)^2(x + 2)$$

Worked Solutions

Polynomial Theorems

Q095 MCQ

Factorise $2x^4 - 5x^3 - 20x^2 + 20x + 48$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 4)(x - 2)(x + 2)(2x + 3)$.

Correct option: D**Final answer**

$$(x - 4)(x - 2)(x + 2)(2x + 3)$$

Q096 MCQ

Factorise $x^4 - 4x^3 - 4x^2 + 13x - 6$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)(x + 2)(x^2 - 5x + 3)$.

Correct option: A**Final answer**

$$(x - 1)(x + 2)(x^2 - 5x + 3)$$

Worked Solutions

Polynomial Theorems

Q097 MCQ

Factorise $x^4 - 4x^3 + 3x^2 + 2x - 6$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 3)(x + 1)(x^2 - 2x + 2)$.

Correct option: B**Final answer**

$$(x - 3)(x + 1)(x^2 - 2x + 2)$$

Q098 MCQ

Factorise $x^4 - x^3 + x^2 + x - 2$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)(x + 1)(x^2 - x + 2)$.

Correct option: C**Final answer**

$$(x - 1)(x + 1)(x^2 - x + 2)$$

Worked Solutions

Polynomial Theorems

Q099 Written

Factorise $2x^3 - 3x^2 - x - 2$.**Solution**

1. $P(2) = 0$; *quotient* $2x^2 + x + 1$ *has discriminant* -7
2. Therefore the final answer is $(x - 2)(2x^2 + x + 1)$.

Final answer

$$(x - 2)(2x^2 + x + 1)$$

Q100 Written

Factorise $2x^3 - 7x^2 + 3x + 6$.**Solution**

1. Test divisor 2 and divide
2. Quadratic quotient remains
3. Therefore the final answer is $(x - 2)(2x^2 - 3x - 3)$.

Final answer

$$(x - 2)(2x^2 - 3x - 3)$$

Worked Solutions

Polynomial Theorems

Q101 Written

Factorise $x^3 + 4x^2 + 2x - 4$.**Solution**

1. Test divisor -2 and divide
2. Quadratic quotient remains
3. Therefore the final answer is $(x + 2)(x^2 + 2x - 2)$.

Final answer

$$(x + 2)(x^2 + 2x - 2)$$

Q102 Written

Factorise $x^3 - 3x^2 + 2$.**Solution**

1. Test divisor 1 and divide
2. Quadratic quotient remains
3. Therefore the final answer is $(x - 1)(x^2 - 2x - 2)$.

Final answer

$$(x - 1)(x^2 - 2x - 2)$$

Worked Solutions

Polynomial Theorems

Q103 MCQ

Factorise $2x^3 - 3x^2 - x - 2$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 2)(2x^2 + x + 1)$.

Correct option: C**Final answer**

$$(x - 2)(2x^2 + x + 1)$$

Q104 MCQ

Factorise $2x^3 - 7x^2 + 3x + 6$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 2)(2x^2 - 3x - 3)$.

Correct option: B**Final answer**

$$(x - 2)(2x^2 - 3x - 3)$$

Worked Solutions

Polynomial Theorems

Q105 MCQ

Factorise $x^3 + 4x^2 + 2x - 4$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x + 2)(x^2 + 2x - 2)$.

Correct option: B**Final answer**

$$(x + 2)(x^2 + 2x - 2)$$

Q106 MCQ

Factorise $x^3 - 3x^2 + 2$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)(x^2 - 2x - 2)$.

Correct option: D**Final answer**

$$(x - 1)(x^2 - 2x - 2)$$

Worked Solutions

Polynomial Theorems

Q107 Challenge

Evaluate the exact remainder when $x^{57} + 57$ is divided by $x + 1$.

Solution

1. *Substitute* $x = -1$
2. $(-1)^{57} + 57 = 56$
3. Therefore the final answer is 56.

Final answer

56

Worked Solutions

Polynomial Theorems

Q108 Challenge

A model is $P(x) = 2x^3 - 3x^2 + ax + 6$. It must be divisible by $2x + 1$. Find a , then state what changes for divisor $x + 2$ and when more than one linear factor can divide a polynomial.

Solution

1. Use the divisor zeros: $P(-1/2) = 0$ and $P(-2) = 0$.
2. Hence $a = 10$ for $2x + 1$, and $a = -11$ for $x + 2$.
3. Each linear factor corresponds to a polynomial root.

Final answer

$$a = 10 \quad \text{for } 2x + 1$$

$$a = -11 \quad \text{for } x + 2$$

linear factors $\leftrightarrow$ roots

Worked Solutions

Polynomial Theorems

Q109 Challenge

A polynomial $P(x)$ is divisible by $x - 1$ and leaves remainder -4 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$.

Solution

1. $a + b = 0$ and $-3a + b = -4$
2. $R = x - 1$
3. Therefore the final answer is $x - 1$.

Final answer $x - 1$

Worked Solutions

Polynomial Theorems

Q110 Challenge

The polynomial $P(x) = 4x^4 - 4x^3 - 11x^2 + 12x - 3$ has a horizontal intercept at $x = 1/2$ with multiplicity 2. Find the other intercepts.

Solution

1. $P = (2x - 1)^2(x^2 - 3)$
2. Remaining roots are $+ / - \sqrt{3}$
3. Therefore the final answer is $x = \sqrt{3}, -\sqrt{3}$.

Final answer

$$x = \sqrt{3}, -\sqrt{3}$$

Worked Solutions

Polynomial Theorems

Q111 Challenge

Let $P(x) = x^4 - 3x^3 + ax^2 + bx + 12$. If both $x - 1$ and $x + 2$ are factors, find a, b .

Solution

1. $P(1) = 0$ gives $a + b = -10$
2. $P(-2) = 0$ gives $2a - b = -26$
3. $a = -12, b = 2$
4. Therefore the final answer is $a = -12, b = 2$.

Final answer

$$a = -12, b = 2$$

Worked Solutions

Polynomial Theorems

Q112 Challenge

For the polynomial above with $a = -12, b = 2$, determine all roots of $P(x) = 0$.

Solution

1. $P = (x - 1)(x + 2)(x^2 - 4x - 6)$
2. Solve quadratic : $x = 2 \pm \sqrt{10}$
3. All four roots satisfy $P = 0$
4. Therefore the final answer is $x = 1, -2, 2 \pm \sqrt{10}$.

Final answer

$$x = 1, -2, 2 \pm \sqrt{10}$$

Worked Solutions

Polynomial Theorems

Q113 Challenge

Evaluate the exact remainder when $x^{57} + 57$ is divided by $x + 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is 56.

Correct option: B

Final answer

56

Worked Solutions

Polynomial Theorems

Q114 Challenge

A model is $P(x) = 2x^3 - 3x^2 + ax + 6$. It must be divisible by $2x + 1$. Find a , then state what changes for divisor $x + 2$ and when more than one linear factor can divide a polynomial. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.

$$a = 10 \quad \text{for } 2x + 1$$

2. The correct value is $a = -11$ for $x + 2$.

linear factors $\leftrightarrow$ roots

Correct option: D

Final answer

$$a = 10 \quad \text{for } 2x + 1$$

$$a = -11 \quad \text{for } x + 2$$

linear factors $\leftrightarrow$ roots

Worked Solutions

Polynomial Theorems

Q115 Challenge

A polynomial $P(x)$ is divisible by $x - 1$ and leaves remainder -4 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $x - 1$.

Correct option: C

Final answer

$x - 1$

Worked Solutions

Polynomial Theorems

Q116 Challenge

The polynomial $P(x) = 4x^4 - 4x^3 - 11x^2 + 12x - 3$ has a horizontal intercept at $x = 1/2$ with multiplicity 2. Find the other intercepts. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $x = \sqrt{3}, -\sqrt{3}$.

Correct option: A

Final answer

$$x = \sqrt{3}, -\sqrt{3}$$

Worked Solutions

Polynomial Theorems

Q117 Challenge

Let $P(x) = x^4 - 3x^3 + ax^2 + bx + 12$. If both $x - 1$ and $x + 2$ are factors, find a, b Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $a = -12, b = 2$.

Correct option: C

Final answer

$$a = -12, b = 2$$

Worked Solutions

Polynomial Theorems

Q118 Challenge

For the polynomial above with $a = -12, b = 2$, determine all roots of $P(x) = 0$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $x = 1, -2, 2 \pm \sqrt{10}$.

Correct option: A

Final answer

$$x = 1, -2, 2 \pm \sqrt{10}$$

Worked Solutions

Polynomial Theorems

Quiz MCQ 1 [Concept Quiz · MCQ](#)

Find the remainder when $P(x) = x^3 - 2x + 5$ is divided by $x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is 9.

Correct option: C

Final answer

9

Quiz MCQ 2 [Concept Quiz · MCQ](#)

If $P(x)$ is divisible by $x + 1$, which statement is true? Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $P(-1) = 0$.

Correct option: C

Final answer

$P(-1) = 0$

Worked Solutions

Polynomial Theorems

Quiz MCQ 3 [Concept Quiz · MCQ](#)

The remainder on division by a quadratic divisor has the form $ax + b$. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $ax + b$.

Correct option: B

Final answer

$$ax + b$$

Quiz MCQ 4 [Concept Quiz · MCQ](#)

Which method records quotient and remainder for division by $x - a$? Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is *Syntheticdivision*.

Correct option: C

Final answer

Syntheticdivision

Worked Solutions

Polynomial Theorems

Quiz WRITTEN 5 Concept Quiz · Written

Find the remainder when $2x^3 + 3x^2 - x + 4$ is divided by $2x + 1$.

Solution

1. Apply the relevant polynomial theorem.
2. Final answer: 4

Final answer

4

Quiz WRITTEN 6 Concept Quiz · Written

Find the remainder $R = ax + b$ when $P(x)$ has values $P(1) = 3$ and $P(-2) = -6$.

Solution

1. Apply the relevant polynomial theorem.
2. Final answer: $R = 3x$

Final answer

$R = 3x$

Worked Solutions

Polynomial Theorems

Quiz WRITTEN 7 Concept Quiz · Written**Factorise** $x^3 - 6x^2 + 11x - 6$.**Solution**

1. Apply the relevant polynomial theorem.
2. Final answer: $(x - 1)(x - 2)(x - 3)$

Final answer

$$(x - 1)(x - 2)(x - 3)$$

Quiz WRITTEN 8 Concept Quiz · Written**If** $x - 1$ **and** $x + 2$ **are factors of** $x^4 - 3x^3 + ax^2 + bx + 12$, **find** a, b .**Solution**

1. Apply the relevant polynomial theorem.
2. Final answer: $a = -12, b = 2$

Final answer

$$a = -12, b = 2$$

Answer key

Use the question number shown on each page.

Q001 2	Q027 A	Q053 $Q = 3x^3 - x^2 + 2x - 6, R = 0$	Q079 C
Q002 6	Q028 A	Q054 D	Q080 D
Q003 C	Q029 A	Q055 $(x - 1)(x + 1)(x + 2)$	Q081 $(x - 2)^2(x^2 - 2x - 4)$
Q004 C	Q030 A	Q056 $(x - 1)(x - 3)(x + 2)$	Q082 $(x - 3)(x - 2)(x + 1)(x + 7)$
Q005 8	Q031 $3x - 2$	Q057 $(x + 2)^2(x - 5)$	Q083 $(x - 1)^2(x^2 - 2x + 3)$
Q006 $-\frac{1}{8}$	Q032 x	Q058 $(x - 2)(2x + 1)(x + 1)$	Q084 $(x - 4)(x - 3)(x - 2)(x - 1)$
Q007 0	Q033 $3x - 1$	Q059 $(x - 3)(x + 1)(x + 2)$	Q085 $(x - 5)(x - 1)^2(x + 2)$
Q008 4	Q034 $7x - 12$	Q060 $(x - 1)(x + 2)(x + 3)$	Q086 $(x - 4)(x - 2)(x + 2)(2x + 3)$
Q009 D	Q035 B	Q061 $(x + 1)(x + 2)(x + 3)$	Q087 $(x - 1)(x + 2)(x^2 - 5x + 3)$
Q010 A	Q036 B	Q062 $(x - 1)^2(x + 2)$	Q088 $(x - 3)(x + 1)(x^2 - 2x + 2)$
Q011 D	Q037 A	Q063 $(x - 3)(x + 2)^2$	Q089 $(x - 1)(x + 1)(x^2 - x + 2)$
Q012 A	Q038 D	Q064 $(x - 5)(x + 1)^2$	Q090 D
Q013 $a = 13/3$	Q039 $x^2 + 3x - 4$	Q065 $(x - 2)(x + 1)(2x - 1)$	Q091 A
Q014 $a = -4$	Q040 A	Q066 $(x - 1)(x + 2)(3x - 2)$	Q092 D
Q015 $a = -4$	Q041 $Q = x^2 - 2x + 9, R = -7$	Q067 $(x + 1)^2(2x + 3)$	Q093 D
Q016 $a = -16$	Q042 A	Q068 D	Q094 C
Q017 B	Q043 $Q = x^2 + x + 6, R = 0$	Q069 D	Q095 D
Q018 B	Q044 $Q = 3x^2 + 4x + 9, R = 26$	Q070 C	Q096 A
Q019 C	Q045 $Q = x^2 + 2x - 1, R = 4$	Q071 B	Q097 B
Q020 A	Q046 $Q = x^2 + 3x + 2, R = -1$	Q072 B	Q098 C
Q021 $a = -4$	Q047 $Q = 4x + 2, R = 4$	Q073 C	Q099 $(x - 2)(2x^2 + x + 1)$
Q022 A	Q048 B	Q074 B	Q100 $(x - 2)(2x^2 - 3x - 3)$
Q023 $2x + 1$	Q049 D	Q075 D	Q101 $(x + 2)(x^2 + 2x - 2)$
Q024 $-10x + 13$	Q050 D	Q076 D	Q102 $(x - 1)(x^2 - 2x - 2)$
Q025 $2x + 3$	Q051 A	Q077 D	Q103 C
Q026 $-15x + 19$	Q052 B	Q078 A	Q104 B

Q105 B**Q106** D**Q107** 56

$$a = 10 \quad \text{for } 2x + 1$$

Q108 $a = -11$ for $x + 2$ linear factors $\leftrightarrow$ roots**Q109** $x - 1$

Q110 $x = \sqrt{3}, -\sqrt{3}$

Q111 $a = -12, b = 2$

Q112 $x = 1, -2, 2 \pm \sqrt{10}$

Q113 B**Q114** D**Q115** C**Q116** A**Q117** C**Q118** A**Quiz MCQ 1** C**Quiz MCQ 2** C**Quiz MCQ 3** B**Quiz MCQ 4** C**Quiz WRITTEN 5** 4**Quiz WRITTEN 6** $R = 3x$ **Quiz WRITTEN 7** $(x - 1)(x - 2)(x - 3)$ **Quiz WRITTEN 8** $a = -12, b = 2$

Concept 4 | Higher-Degree Equations

English Student Edition • Focused formulas and guided practice

Formula opener

Factor theorem

$$P(a) = 0 \Rightarrow x - a \text{ is a factor}$$

Degree

$$\deg(PQ) = \deg P + \deg Q$$

Conjugate roots

$$a + bi \Rightarrow a - bi$$

Worked Solutions

Higher-Degree Equations

Q001 Written

$$x^3 - 8 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$.

Final answer

$$x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$$

Q002 Written

$$x^3 = -1$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$.

Final answer

$$x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$$

Worked Solutions

Higher-Degree Equations

Q003 Written

$$x^3 = 27$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$.

Final answer

$$x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$$

Q004 Written

$$x^3 = -125$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$.

Final answer

$$x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$$

Worked Solutions

Higher-Degree Equations

Q005 Written

$$8x^3 = 1$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$.

Final answer

$$x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$$

Q006 MCQ

$$x^3 - 8 = 0$$
 Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$.

Correct option: B

Final answer

$$x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$$

Worked Solutions

Higher-Degree Equations

Q007 MCQ

 $x^3 = -1$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$.

Correct option: D

Final answer

$$x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$$

Q008 MCQ

 $x^3 = 27$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$.

Correct option: B

Final answer

$$x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$$

Worked Solutions

Higher-Degree Equations

Q009 MCQ

 $x^3 = -125$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$.

Correct option: A

Final answer

$$x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$$

Q010 MCQ

 $8x^3 = 1$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$.

Correct option: C

Final answer

$$x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$$

Worked Solutions

Higher-Degree Equations

Q011 Written

$$x^3 - 2x^2 - 5x + 6 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, 1, 3$.

Final answer

$$x = -2, 1, 3$$

Q012 Written

$$3x^3 + x^2 - 8x + 4 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, 2/3, 1$.

Final answer

$$x = -2, 2/3, 1$$

Worked Solutions

Higher-Degree Equations

Q013 Written

$$2x^3 - 3x^2 - x - 2 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$.

Final answer

$$x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$$

Q014 Written

$$x^3 - x^2 - 16x - 20 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2(\text{repeated}), 5$.

Final answer

$$x = -2(\text{repeated}), 5$$

Worked Solutions

Higher-Degree Equations

Q015 Written

$$x^3 + 6x^2 + 11x + 6 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, -2, -1$.

Final answer

$$x = -3, -2, -1$$

Q016 Written

$$x^3 + 3x^2 - 4x - 12 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, -2, 2$.

Final answer

$$x = -3, -2, 2$$

Worked Solutions

Higher-Degree Equations

Q017 Written

$$3x^3 - 2x^2 - 3x + 2 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1, 2/3, 1$.

Final answer

$$x = -1, 2/3, 1$$

Q018 Written

$$x^3 - x^2 - 4 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$.

Final answer

$$x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$$

Worked Solutions

Higher-Degree Equations

Q019 Written

$$x^3 + 4x^2 + 2x - 4 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$.

Final answer

$$x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$$

Q020 Written

$$x^3 - 4x^2 + 9x - 10 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, 1 - 2i, 1 + 2i$.

Final answer

$$x = 2, 1 - 2i, 1 + 2i$$

Worked Solutions

Higher-Degree Equations

Q021 Written

$$x^3 + x^2 - 8x - 12 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2$ (repeated), 3.

Final answer

$$x = -2(\text{repeated}), 3$$

Q022 Written

$$2x^3 + 7x^2 + 8x + 3 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1$ (repeated), $-3/2$.

Final answer

$$x = -1(\text{repeated}), -3/2$$

Worked Solutions

Higher-Degree Equations

Q023 Written

$$x^3 + 6x^2 + 12x + 8 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2$ (triple).

Final answer

$$x = -2(\text{triple})$$

Q024 MCQ

$$x^3 - 2x^2 - 5x + 6 = 0$$
 Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -2, 1, 3$.

Correct option: C

Final answer

$$x = -2, 1, 3$$

Worked Solutions

Higher-Degree Equations

Q025 MCQ

 $3x^3 + x^2 - 8x + 4 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -2, 2/3, 1$.

Correct option: D

Final answer

$$x = -2, 2/3, 1$$

Q026 MCQ

 $2x^3 - 3x^2 - x - 2 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$.

Correct option: D

Final answer

$$x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$$

Worked Solutions

Higher-Degree Equations

Q027 MCQ

 $x^3 - x^2 - 16x - 20 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -2$ (repeated), 5.

Correct option: B

Final answer $x = -2$ (repeated), 5

Q028 MCQ

 $x^3 + 6x^2 + 11x + 6 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -3, -2, -1$.

Correct option: C

Final answer $x = -3, -2, -1$

Worked Solutions

Higher-Degree Equations

Q029 MCQ

 $x^3 + 3x^2 - 4x - 12 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -3, -2, 2$.

Correct option: D

Final answer

$$x = -3, -2, 2$$

Q030 MCQ

 $3x^3 - 2x^2 - 3x + 2 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -1, 2/3, 1$.

Correct option: D

Final answer

$$x = -1, 2/3, 1$$

Worked Solutions

Higher-Degree Equations

Q031 MCQ

 $x^3 - x^2 - 4 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$.

Correct option: A

Final answer

$$x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$$

Worked Solutions

Higher-Degree Equations

Q032 MCQ

 $x^3 + 4x^2 + 2x - 4 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$.

Correct option: D

Final answer

$$x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$$

Worked Solutions

Higher-Degree Equations

Q033 MCQ

 $x^3 - 4x^2 + 9x - 10 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 2, 1 - 2i, 1 + 2i$.

Correct option: D

Final answer

$$x = 2, 1 - 2i, 1 + 2i$$

Q034 MCQ

 $x^3 + x^2 - 8x - 12 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -2(\text{repeated}), 3$.

Correct option: D

Final answer

$$x = -2(\text{repeated}), 3$$

Worked Solutions

Higher-Degree Equations

Q035 MCQ

 $2x^3 + 7x^2 + 8x + 3 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -1$ (repeated), $-3/2$.

Correct option: B**Final answer**

$$x = -1(\text{repeated}), -3/2$$

Q036 MCQ

 $x^3 + 6x^2 + 12x + 8 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -2$ (triple).

Correct option: D**Final answer**

$$x = -2(\text{triple})$$

Worked Solutions

Higher-Degree Equations

Q037 Written

$$x^4 - 7x^2 - 18 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, 3, -\sqrt{2}i, \sqrt{2}i$.

Final answer

$$x = -3, 3, -\sqrt{2}i, \sqrt{2}i$$

Q038 Written

$$x^4 - 5x^2 + 4 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, -1, 1, 2$.

Final answer

$$x = -2, -1, 1, 2$$

Worked Solutions

Higher-Degree Equations

Q039 Written

$$x^4 - 5x^2 - 36 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, 3, -2i, 2i$.

Final answer

$$x = -3, 3, -2i, 2i$$

Q040 Written

$$x^4 - 6x^2 - 27 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, 3, -\sqrt{3}i, \sqrt{3}i$.

Final answer

$$x = -3, 3, -\sqrt{3}i, \sqrt{3}i$$

Worked Solutions

Higher-Degree Equations

Q041 Written

$$x^4 - x^2 - 2 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -\sqrt{2}, \sqrt{2}, -i, i$.

Final answer

$$x = -\sqrt{2}, \sqrt{2}, -i, i$$

Q042 MCQ

$$x^4 - 7x^2 - 18 = 0$$
 Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -3, 3, -\sqrt{2}i, \sqrt{2}i$.

Correct option: D

Final answer

$$x = -3, 3, -\sqrt{2}i, \sqrt{2}i$$

Worked Solutions

Higher-Degree Equations

Q043 MCQ

 $x^4 - 5x^2 + 4 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -2, -1, 1, 2$.

Correct option: D

Final answer

$$x = -2, -1, 1, 2$$

Q044 MCQ

 $x^4 - 5x^2 - 36 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -3, 3, -2i, 2i$.

Correct option: C

Final answer

$$x = -3, 3, -2i, 2i$$

Worked Solutions

Higher-Degree Equations

Q045 MCQ

 $x^4 - 6x^2 - 27 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -3, 3, -\sqrt{3}i, \sqrt{3}i$.

Correct option: C

Final answer

$$x = -3, 3, -\sqrt{3}i, \sqrt{3}i$$

Q046 MCQ

 $x^4 - x^2 - 2 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -\sqrt{2}, \sqrt{2}, -i, i$.

Correct option: B

Final answer

$$x = -\sqrt{2}, \sqrt{2}, -i, i$$

Worked Solutions

Higher-Degree Equations

Q047 Written

$$x^4 - 6x^3 + 8x^2 + 8x - 16 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$.

Final answer

$$x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$$

Q048 Written

$$x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 1, 2, 3, 4$.

Final answer

$$x = 1, 2, 3, 4$$

Worked Solutions

Higher-Degree Equations

Q049 Written

$$x^4 - 9x^3 + 29x^2 - 39x + 18 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 1, 2, 3(\text{repeated})$.

Final answer

$$x = 1, 2, 3(\text{repeated})$$

Q050 Written

$$x^4 - 4x^3 + 3x^2 + 2x - 6 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1, 3, 1 - i, 1 + i$.

Final answer

$$x = -1, 3, 1 - i, 1 + i$$

Worked Solutions

Higher-Degree Equations

Q051 MCQ

 $x^4 - 6x^3 + 8x^2 + 8x - 16 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 2$ (repeated), $1 - \sqrt{5}$, $1 + \sqrt{5}$.

Correct option: B

Final answer

$$x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$$

Worked Solutions

Higher-Degree Equations

Q052 MCQ

 $x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 1, 2, 3, 4$.

Correct option: A

Final answer $x = 1, 2, 3, 4$

Q053 MCQ

 $x^4 - 9x^3 + 29x^2 - 39x + 18 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 1, 2, 3(\text{repeated})$.

Correct option: C

Final answer $x = 1, 2, 3(\text{repeated})$

Worked Solutions

Higher-Degree Equations

Q054 MCQ

 $x^4 - 4x^3 + 3x^2 + 2x - 6 = 0$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -1, 3, 1 - i, 1 + i$.

Correct option: C

Final answer

$$x = -1, 3, 1 - i, 1 + i$$

Q055 Written

 $x^3 + 4x^2 + ax + b = 0$ has roots -3 and 1; find a,b and the other root.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = 1, b = -6$; other roots: -2 .

Final answer

$$a = 1, b = -6; \text{ other roots: } -2$$

Worked Solutions

Higher-Degree Equations

Q056 Written

$x^3 - ax^2 - x + b = 0$ has roots -1 and 2; find a, b and the other root.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = 2, b = 2$; other roots: 1.

Final answer

$a = 2, b = 2$; other roots: 1

Q057 Written

$x^3 + ax^2 + bx - 8 = 0$ has roots 1 and 2; find a, b and the other root.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -7, b = 14$; other roots: 4.

Final answer

$a = -7, b = 14$; other roots: 4

Worked Solutions

Higher-Degree Equations

Q058 MCQ

$x^3 + 4x^2 + ax + b = 0$ has roots -3 and 1; find a,b and the other root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = 1, b = -6$; other roots: -2 .

Correct option: C

Final answer

$a = 1, b = -6$; other roots: -2

Worked Solutions

Higher-Degree Equations

Q059 MCQ

$x^3 - ax^2 - x + b = 0$ has roots -1 and 2; find a, b and the other root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = 2, b = 2$; other roots: 1.

Correct option: B

Final answer

$a = 2, b = 2$; other roots: 1

Worked Solutions

Higher-Degree Equations

Q060 MCQ

$x^3 + ax^2 + bx - 8 = 0$ has roots 1 and 2; find a, b and the other root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -7, b = 14$; other roots: 4.

Correct option: C

Final answer

$a = -7, b = 14$; other roots: 4

Worked Solutions

Higher-Degree Equations

Q061 Written

$x^3 + ax^2 + bx + 5 = 0$ has root $2+i$; find real a, b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -3, b = 1$; other roots: $-1, 2 - i$.

Final answer

$a = -3, b = 1$; other roots: $-1, 2 - i$

Q062 Written

$x^3 + ax^2 + bx - 4 = 0$ has root $1+i$; find a, b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -4, b = 6$; other roots: $1 - i, 2$.

Final answer

$a = -4, b = 6$; other roots: $1 - i, 2$

Worked Solutions

Higher-Degree Equations

Q063 Written

$x^3 + ax^2 + bx - 20 = 0$ has root $1 - 3i$; find real a, b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -4, b = 14$; other roots: $1 + 3i, 2$.

Final answer

$a = -4, b = 14$; other roots: $1 + 3i, 2$

Q064 Written

$x^3 + ax^2 + bx + 10 = 0$ has root $1 + 2i$; find real a, b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = 0, b = 1$; other roots: $1 - 2i, -2$.

Final answer

$a = 0, b = 1$; other roots: $1 - 2i, -2$

Worked Solutions

Higher-Degree Equations

Q065 MCQ

$x^3 + ax^2 + bx + 5 = 0$ has root $2+i$; find real a, b and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -3, b = 1$; other roots: $-1, 2 - i$.

Correct option: A

Final answer

$a = -3, b = 1$; other roots: $-1, 2 - i$

Q066 MCQ

$x^3 + ax^2 + bx - 4 = 0$ has root $1+i$; find a, b and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -4, b = 6$; other roots: $1 - i, 2$.

Correct option: B

Final answer

$a = -4, b = 6$; other roots: $1 - i, 2$

Worked Solutions

Higher-Degree Equations

Q067 MCQ

$x^3 + ax^2 + bx - 20 = 0$ has root $1 - 3i$; find real a, b and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -4, b = 14$; other roots: $1 + 3i, 2$.

Correct option: D

Final answer

$a = -4, b = 14$; other roots: $1 + 3i, 2$

Q068 MCQ

$x^3 + ax^2 + bx + 10 = 0$ has root $1 + 2i$; find real a, b and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = 0, b = 1$; other roots: $1 - 2i, -2$.

Correct option: B

Final answer

$a = 0, b = 1$; other roots: $1 - 2i, -2$

Worked Solutions

Higher-Degree Equations

Q069 Written

$x^3 - kx^2 + (3k - 1)x - 24 = 0$ has root 2; find k and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $k = 9$; other roots: 3, 4.

Final answer

$k = 9$; other roots: 3, 4

Q070 MCQ

$x^3 - kx^2 + (3k - 1)x - 24 = 0$ has root 2; find k and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $k = 9$; other roots: 3, 4.

Correct option: B

Final answer

$k = 9$; other roots: 3, 4

Worked Solutions

Higher-Degree Equations

Q071 Written

For an imaginary cube root of unity w , evaluate w^9 .**Solution**

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer

1

Q072 Written

For an imaginary cube root of unity w , evaluate w^{2010} .**Solution**

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer

1

Worked Solutions

Higher-Degree Equations

Q073 **Written****For an imaginary cube root of unity w , evaluate w^3 .****Solution**

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer

1

Q074 **MCQ****For an imaginary cube root of unity w , evaluate w^9 Select the correct answer:****Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is 1.

Correct option: A**Final answer**

1

Worked Solutions

Higher-Degree Equations

Q075 MCQ

For an imaginary cube root of unity w , evaluate w^{2010} Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is 1.

Correct option: B**Final answer**

1

Q076 MCQ

For an imaginary cube root of unity w , evaluate w^3 Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is 1.

Correct option: D**Final answer**

1

Worked Solutions

Higher-Degree Equations

Q077 Written

For an imaginary cube root of unity w , evaluate $w^4 + w^5$.**Solution**

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is -1 .

Final answer -1

Q078 Written

For an imaginary cube root of unity w , evaluate $w^2 + w$.**Solution**

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is -1 .

Final answer -1

Worked Solutions

Higher-Degree Equations

Q079 **Written**

For an imaginary cube root of unity w , evaluate $w^2 + w^4 + w^6$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 0.

Final answer

0

Q080 **MCQ**

For an imaginary cube root of unity w , evaluate $w^4 + w^5$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is -1 .

Correct option: A

Final answer -1

Worked Solutions

Higher-Degree Equations

Q081 MCQ

For an imaginary cube root of unity w , evaluate $w^2 + w$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is -1 .

Correct option: B**Final answer** -1

Q082 MCQ

For an imaginary cube root of unity w , evaluate $w^2 + w^4 + w^6$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is 0 .

Correct option: D**Final answer** 0

Worked Solutions

Higher-Degree Equations

Q083 Written

For an imaginary cube root of unity w , evaluate $w + 1/w$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is -1 .

Final answer -1

Q084 Written

For an imaginary cube root of unity w , evaluate $(1 + w)(1 + w^2)$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer 1

Worked Solutions

Higher-Degree Equations

Q085 Written

For an imaginary cube root of unity w , evaluate $1/w + 1/w^2 + 1$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 0.

Final answer

0

Q086 MCQ

For an imaginary cube root of unity w , evaluate $w + 1/w$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is -1 .

Correct option: D

Final answer -1

Worked Solutions

Higher-Degree Equations

Q087 MCQ

For an imaginary cube root of unity w , evaluate $(1 + w)(1 + w^2)$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is 1.

Correct option: C

Final answer

1

Q088 MCQ

For an imaginary cube root of unity w , evaluate $1/w + 1/w^2 + 1$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is 0.

Correct option: D

Final answer

0

Worked Solutions

Higher-Degree Equations

Q089 Written

$x^3 - 2x^2 + (a - 3)x + a = 0$ Find all real parameter values a producing a repeated root.

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = -4$ or $9/4$.

Final answer

$a = -4$ or $9/4$

Q090 Written

$x^3 - x^2 + (a - 2)x + a = 0$ Find all real parameter values a producing a repeated root.

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = -3$ or 1 .

Final answer

$a = -3$ or 1

Worked Solutions

Higher-Degree Equations

Q091 Written

$x^3 - 7x^2 + (a + 6)x - a = 0$ Find all real parameter values a producing a repeated root.

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = 5$ or 9 .

Final answer

$a = 5$ or 9

Q092 Written

$x^3 - (a + 1)x^2 + (3a + 5)x - 2a - 5 = 0$ Find all real parameter values a producing a repeated root.

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = -6, -2$, or 10 .

Final answer

$a = -6, -2$, or 10

Worked Solutions

Higher-Degree Equations

Q093 Written

$x^3 + (2a - 1)x^2 - 3(a - 2)x + a - 6 = 0$ Find all real parameter values a producing a repeated root.

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = -7, -3$, or 2 .

Final answer $a = -7, -3$, or 2

Q094 MCQ

$x^3 - 2x^2 + (a - 3)x + a = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -4$ or $9/4$.

Correct option: D

Final answer $a = -4$ or $9/4$

Worked Solutions

Higher-Degree Equations

Q095 MCQ

$x^3 - x^2 + (a - 2)x + a = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -3$ or 1 .

Correct option: D

Final answer

$a = -3$ or 1

Q096 MCQ

$x^3 - 7x^2 + (a + 6)x - a = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = 5$ or 9 .

Correct option: B

Final answer

$a = 5$ or 9

Worked Solutions

Higher-Degree Equations

Q097 MCQ

$x^3 - (a + 1)x^2 + (3a + 5)x - 2a - 5 = 0$ Find all real parameter values a producing a repeated root. Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -6, -2$, or 10 .

Correct option: A**Final answer** $a = -6, -2$, or 10

Q098 MCQ

$x^3 + (2a - 1)x^2 - 3(a - 2)x + a - 6 = 0$ Find all real parameter values a producing a repeated root. Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -7, -3$, or 2 .

Correct option: A**Final answer** $a = -7, -3$, or 2

Worked Solutions

Higher-Degree Equations

Q099 Challenge

$x^3 + ax + b = 0$ has root $1 + \sqrt{2}i$; find real a, b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$.

Final answer

$a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$

Worked Solutions

Higher-Degree Equations

Q100 Challenge

For an imaginary cube root of unity w , evaluate $(1 - w + w^2)(1 + w - w^2)$ without direct expansion.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 4.

Final answer

4

Worked Solutions

Higher-Degree Equations

Q101 Challenge

$x^3 + (a - 1)x^2 + (a - 3)x - 2a + 3 = 0$ Find all real parameter values a producing a repeated root.

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = 2/3, 2$, or 6 .

Final answer

$a = 2/3, 2$, or 6

Worked Solutions

Higher-Degree Equations

Q102 Challenge

Write the balance equation: inflow $x^4 + x^2 + x$ equals outflow $x^3 + 2$.

Solution

1. Set inflow equal to outflow: $(x^4 + x^2 + x = x^3 + 2)$.
2. Move all terms to one side.
3. Therefore the final answer is $x^4 - x^3 + x^2 + x - 2 = 0$.

Final answer

$$x^4 - x^3 + x^2 + x - 2 = 0$$

Worked Solutions

Higher-Degree Equations

Q103 Challenge

Solve $x^4 - x^3 + x^2 + x - 2 = 0$ **for the stable pressure levels.**

Solution

1. Factor $(x^4 - x^3 + x^2 + x - 2)$ and solve the remaining quadratic.
2. Apply the stated physical restriction ($x \geq 0$) only after listing the algebraic roots.
3. Therefore the final answer is $x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$.

Final answer

$$x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$$

Worked Solutions

Higher-Degree Equations

Q104 Challenge

Classify the stable pressure solutions as real or complex.

Solution

1. Separate the real roots from the non-real conjugate pair.

Both types occur: real $x = -1, 1$

2. Therefore the final answer is non-real conjugates $x = (1 \pm \sqrt{7}i)/2$.

if $x \geq 0 : x = 1$

Final answer

Both types occur: real $x = -1, 1$

non-real conjugates $x = (1 \pm \sqrt{7}i)/2$

if $x \geq 0 : x = 1$

Worked Solutions

Higher-Degree Equations

Q105 Challenge

$x^3 + ax + b = 0$ has root $1 + \sqrt{2}i$; find real a, b and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$.

Correct option: C

Final answer

$a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$

Worked Solutions

Higher-Degree Equations

Q106 Challenge

For an imaginary cube root of unity w , evaluate $(1 - w + w^2)(1 + w - w^2)$ without direct expansion. Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is 4.

Correct option: D

Final answer

4

Worked Solutions

Higher-Degree Equations

Q107 Challenge

$x^3 + (a - 1)x^2 + (a - 3)x - 2a + 3 = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = 2/3, 2$, or 6 .

Correct option: D

Final answer

$a = 2/3, 2$, or 6

Worked Solutions

Higher-Degree Equations

Q108 Challenge

Write the balance equation: inflow $x^4 + x^2 + x$ equals outflow $x^3 + 2$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x^4 - x^3 + x^2 + x - 2 = 0$.

Correct option: A

Final answer

$$x^4 - x^3 + x^2 + x - 2 = 0$$

Worked Solutions

Higher-Degree Equations

Q109 Challenge

Solve $x^4 - x^3 + x^2 + x - 2 = 0$ **for the stable pressure levels Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$.

Correct option: D

Final answer

$$x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$$

Worked Solutions

Higher-Degree Equations

Q110 Challenge

Classify the stable pressure solutions as real or complex Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.

Both types occur: real $x = -1, 1$

2. The correct value is non-real conjugates $x = (1 \pm \sqrt{7}i)/2$.

if $x \geq 0 : x = 1$

Correct option: B

Final answer

Both types occur: real $x = -1, 1$

non-real conjugates $x = (1 \pm \sqrt{7}i)/2$

if $x \geq 0 : x = 1$

Worked Solutions

Higher-Degree Equations

Quiz MCQ 1 **Concept Quiz · MCQ**

Solve $x^3 - 8 = 0$. Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$.

Correct option: D

Final answer

$$x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$$

Quiz MCQ 2 **Concept Quiz · MCQ**

Are $1 + 2i$ and $1 - 2i$ roots of $x^3 - 2x^2 + 5x - 6 = 0$? Which root must also occur? Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $1 - 2i$.

Correct option: D

Final answer

$$1 - 2i$$

Worked Solutions

Higher-Degree Equations

Quiz MCQ 3 **Concept Quiz · MCQ**

For an imaginary cube root of unity w , evaluate $w^2 + w$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is -1 .

Correct option: B

Final answer

-1

Quiz MCQ 4 **Concept Quiz · MCQ**

For the remaining quadratic factor, a repeated root requires Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $D = 0$.

Correct option: B

Final answer

$D = 0$

Worked Solutions

Higher-Degree Equations

Quiz WRITTEN 5 Concept Quiz · Written

$$\text{Solve } x^4 - 5x^2 + 4 = 0.$$

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: $x = -2, -1, 1, 2$

Final answer

$$x = -2, -1, 1, 2$$

Quiz WRITTEN 6 Concept Quiz · Written

$$\text{For } x^3 + ax^2 + bx + 10 = 0 \text{ with root } 1 + 2i, \text{ find } a, b \text{ and the other roots.}$$

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: $a = 0, b = 1$; other roots: $1 - 2i, -2$

Final answer

$$a = 0, b = 1; \text{ other roots: } 1 - 2i, -2$$

Worked Solutions

Higher-Degree Equations

Quiz WRITTEN 7 Concept Quiz · Written

For an imaginary cube root of unity w , evaluate $(1 + w)(1 + w^2)$.

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: 1

Final answer

1

Quiz WRITTEN 8 Concept Quiz · Written

Find all real a for $x^3 - 2x^2 + (a - 3)x + a = 0$ to have a repeated root.

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: $a = -4$ or $9/4$

Final answer

$a = -4$ or $9/4$

Answer key

Use the question number shown on each page.

Q001 $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$	Q027 B	Q053 C	Q079 0
Q002 $x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$	Q028 C	Q054 C	Q080 A
Q003 $x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$	Q029 D	Q055 $a = 1, b = -6$; other roots: -2	Q081 B
Q004 $x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$	Q030 D	Q056 $a = 2, b = 2$; other roots: 1	Q082 D
Q005 $x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$	Q031 A	Q057 $a = -7, b = 14$; other roots: 4	Q083 -1
Q006 B	Q032 D	Q058 C	Q084 1
Q007 D	Q033 D	Q059 B	Q085 0
Q008 B	Q034 D	Q060 C	Q086 D
Q009 A	Q035 B	Q061 $a = -3, b = 1$; other roots: $-1, 2 - i$	Q087 C
Q010 C	Q036 D	Q062 $a = -4, b = 6$; other roots: $1 - i, 2$	Q088 D
Q011 $x = -2, 1, 3$	Q037 $x = -3, 3, -\sqrt{2}i, \sqrt{2}i$	Q063 $a = -4, b = 14$; other roots: $1 + 3i, 2$	Q089 $a = -4$ or $9/4$
Q012 $x = -2, 2/3, 1$	Q038 $x = -2, -1, 1, 2$	Q064 $a = 0, b = 1$; other roots: $1 - 2i, -2$	Q090 $a = -3$ or 1
Q013 $x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$	Q039 $x = -3, 3, -2i, 2i$	Q065 A	Q091 $a = 5$ or 9
Q014 $x = -2$ (repeated), 5	Q040 $x = -3, 3, -\sqrt{3}i, \sqrt{3}i$	Q066 B	Q092 $a = -6, -2$, or 10
Q015 $x = -3, -2, -1$	Q041 $x = -\sqrt{2}, \sqrt{2}, -i, i$	Q067 D	Q093 $a = -7, -3$, or 2
Q016 $x = -3, -2, 2$	Q042 D	Q068 B	Q094 D
Q017 $x = -1, 2/3, 1$	Q043 D	Q069 $k = 9$; other roots: $3, 4$	Q095 D
Q018 $x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$	Q044 C	Q070 B	Q096 B
Q019 $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$	Q045 C	Q071 1	Q097 A
Q020 $x = 2, 1 - 2i, 1 + 2i$	Q046 B	Q072 1	Q098 A
Q021 $x = -2$ (repeated), 3	Q047 $x = 2$ (repeated), $1 - \sqrt{5}, 1 + \sqrt{5}$	Q073 1	Q099 $a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$
Q022 $x = -1$ (repeated), $-3/2$	Q048 $x = 1, 2, 3, 4$	Q074 A	Q100 4
Q023 $x = -2$ (triple)	Q049 $x = 1, 2, 3$ (repeated)	Q075 B	Q101 $a = 2/3, 2$, or 6
Q024 C	Q050 $x = -1, 3, 1 - i, 1 + i$	Q076 D	Q102 $x^4 - x^3 + x^2 + x - 2 = 0$
Q025 D	Q051 B	Q077 -1	Q103 $x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$
Q026 D	Q052 A	Q078 -1	

Both types occur: real $x = -1, 1$

Q104 non-real conjugates $x = (1 \pm \sqrt{7}i)/2$
if $x \geq 0 : x = 1$

Q105 C

Q106 D

Q107 D

Q108 A

Q109 D

Q110 B

Quiz MCQ 1 D

Quiz MCQ 2 D

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 $x = -2, -1, 1, 2$

Quiz WRITTEN 6 $a = 0, b = 1$; other roots: $1 - 2i, -2$

Quiz WRITTEN 7 1

Quiz WRITTEN 8 $a = -4$ or $9/4$